ESPI



WHERE SENSING MAKES SENSE

OPPORTUNITIES FOR ADVANCING EUROPE'S PRIORITIES THROUGH EARTH OBSERVATION

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TAKEAWAYS & PATH FORWARD

It is well established that data obtained through Earth Observation satellites directly contributes to priority objectives across a number of policy domains. Among these, climate and agriculture policies are benefitting the most from economic and societal impacts generated by public infrastructures, such as Copernicus, and commercial missions alike. Yet, the analysis pursued in this study shows that a number of other policy domains, and potential use cases therein, remain under-addressed. Directly focusing on these can improve the uptake of space solutions, which has been called for across a number of assessments and strategies.[1]

1 EXPANDING IMPACT BEYOND MONITORING FUNCTIONS

Addressing today's policy priorities requires operational solutions that are both ubiquitously available and easily accessible. The primary challenge lies in **moving beyond preventive and post-event monitoring towards real-time full stack architectures** directly supporting economic activity and public services.

2 HARNESSING THE CROSS-POLICY VALUE OF EARTH OBSERVATION

The ability of Earth Observation to deliver **benefits across a diverse range of policy objectives** leveraging the same datasets, remains underused. To unlock greater policy impact, broader diversification and customisation of solutions are needed. This includes adapting successful solutions to other use cases in different domains.

3 UNTAPPED POTENTIAL AND OPPORTUNITIES FOR EUROPEAN POLICIES

All the identified top-priority domains and sectors – Energy, Agrifood, Digital, Peace & Security, Climate Resilience and Insurance – present distinct and varying opportunities for further contribution of Earth Observation solutions. In the **Energy** and **Digital Transition** domains, there remains a **noticeable gap**, coupled with **demonstrated potential** for dedicated Earth Observation solutions to address their core objectives.

4 INTEGRATING SPACE INTO POLICIES AS A DRIVER FOR IMPACT

A virtuous cycle has been identified and should be sustained: the growing number of impactful Earth Observation projects and the advocacy efforts by space actors raise awareness among policymakers on the existence and value of available solutions. In turn, **stronger integration of references to Earth Observation in policies creates opportunities for space actors to develop projects in alignment with policy priorities.**

[1] *A Vision for the European Space Economy, EU, 2026; Accelerating the Use of Space in Europe ("Matosinhos Manifesto") ESA, 2021; Special Report on EU Space Programmes Galileo and Copernicus, ECA, 2021.*

FROM TAKEAWAYS TO POLICY ACTION

This report provides **quantitative outcomes to support the integration of policy priorities into current and future European Earth Observation systems.**

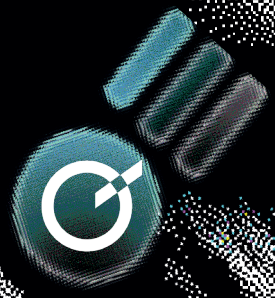
This study and a high-level workshop held at the Royal Society in London in April 2025 informed **A Policy Vision for Earth Observation**. Jointly released by ESPI and ESA, the Vision calls for a new era where the technological & scientific drivers for designing and developing new missions are complemented by a third pillar – directly responding to Europe's policy needs.

It calls on policy and thought leaders to guide and co-shape future Earth Observation programmes, and integrate derived solutions across the entire policy cycle. Where policy needs are most acute, space must follow, especially into uncharted territory.

EXECUTIVE SUMMARY

WHERE SENSING MAKES SENSE

Opportunities for Advancing Europe's Priorities through Earth Observation



Earth Observation systems are increasingly central to achieving a host of public policy objectives. **Efforts to scale and consolidate the impact of Earth Observation solutions in relevant policy areas** should be guided by a sound identification and assessment of existing gaps and opportunities.

This study has been developed to provide evidence on the current level of contribution of Earth Observation solutions to priority policy domains across European countries. It aims to support space actors and policymakers in taking decisions on the use and tailoring of current solutions, as well as the development of new programmes and initiatives to serve policy priorities in Europe.

Over 300 policy documents and 600 downstream projects leveraging Earth Observation data were analysed across six domains: Energy, Agriculture & Food, Digital Transition, Peace & Security, Climate Resilience & Emergency Management, and Insurance. These were defined and selected according to current high-level and short-term policy **priorities of European countries.**

The policy analysis enabled the identification of approximately thirty concrete use cases per domain, followed by an assessment of their prioritisation in national policies and complemented by a separate overview of the insurance market. The collected projects are distributed unevenly across the selected domains and use cases. Specifically, Earth Observation solutions are yet to systematically address objectives in Energy and Digital Transition, where the current number and diversification of solutions remain limited. **Opportunity vectors arise in each of the analysed domains** and are accompanied by several **case studies on selected public administrations, highlighting the tangible benefits of using Earth Observation** in their activities. Benefits include significant cost and time savings in agricultural field inspections and maritime safety and security activities, increased coverage and information reliability for forestry and other land-monitoring actions, and enhanced policy implementation and compliance with EU and national laws.

Two major clusters of policy objectives deserve further investigation: **Diplomatic Action & International Affairs** and **Natural Resource Management**. While most use cases in both areas show **high prioritisation in national policies**, the former community would benefit from a more significant availability of targeted Earth Observation projects, while the latter requires further understanding of the **strategic value and potential for scale** of available solutions.

Additionally, the explicit prevalence of Earth Observation terminology in non-space policies has been analysed, serving as an indicator of the level of awareness regarding the role of Earth Observation among policymakers and as an entry point for space actors to foster uptake and budget allocation to relevant solutions.

The Climate and Agriculture domains stand out in this context, boasting a high number of references to specific Earth Observation programmes such as Copernicus. In contrast, significantly more limited integration of actionable references to Earth Observation is observed in Energy and Security policies and related implementation plans.

Overall, research shows **significant alignment between the number of Earth Observation projects addressing the use cases in a given domain and the number of references to Earth Observation solutions in respective national policies**. The Energy, Climate and Agriculture domains are particularly representative of this correlation.

Going a step beyond, this report outlines **potential opportunities for increasing the use and benefits of Earth Observation** in specific policy areas, as well as highlighting some overarching features of Earth Observation uptake in each of the selected domains.

POLICY-DRIVEN OPPORTUNITIES FOR EARTH OBSERVATION

Main opportunities emerging across selected domains from a comparison between policy priorities and the number of Earth Observation projects currently addressing them

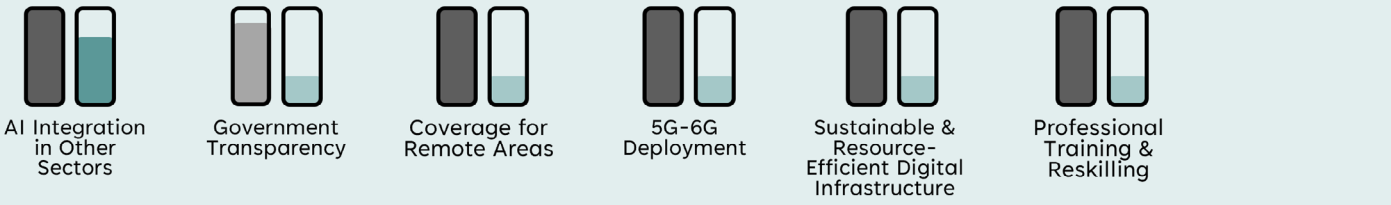
ENERGY



AGRICULTURE



DIGITAL TRANSITION



PEACE & SECURITY (excluding defence-related use cases)

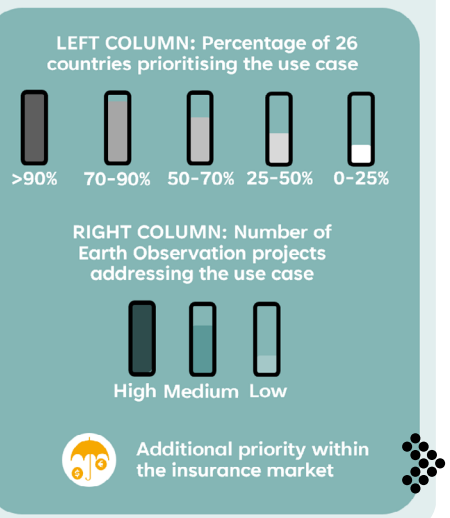


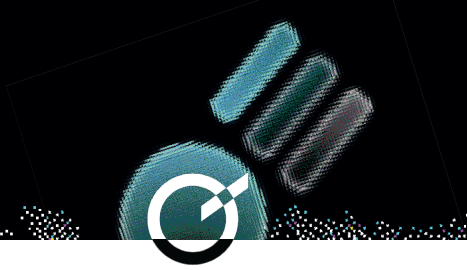
CLIMATE RESILIENCE



INSURANCE (across other selected domains)

- Critical Infrastructure & Essential Systems
- Geological Hazards
- Wars, Riots & Civil Unrest





In **ENERGY**, low alignment between the level of policy ambition and the current contribution of Earth Observation to energy-related objectives is observed. Similarly, explicit references to Earth Observation in energy policies remain very limited. Opportunities lie in the use of EO for the electrification of transport networks, the increase of energy efficiency of buildings & industrial processes, and the deployment of renewables.

In **AGRICULTURE AND FOOD**, Earth Observation solutions are increasingly positioned as enablers of efficient and resilient agricultural practices, while also contributing to sustainable rural development. In this context, services that can be tailored to local realities and scaled by replicability of modular solutions appear to have the highest potential. These could be designed within and beyond broad data platforms, also favouring the integration of in situ data. Opportunities for the use of EO beyond established applications are observed for water quality assurance, management & security of the food value chain, and biodiversity protection & extension.

In **DIGITAL**, Earth Observation increasingly functions as a source of trusted data underpinning the development of applications driving a trusted digital transition. This includes decades of continuous Earth Observation datasets in the public domain that can be integrated into platforms and AI tools, contributing to maximising their impact for industry and public administrations across other sectors and domains. These tools include digital twins, currently mainly focused on urban contexts but with the potential to also boost rural planning and resilience. Specific opportunities can arise from leveraging Earth Observation for enhanced AI applications serving priorities in other domains and sectors, 5G-6G deployment & digital coverage increase for remote areas, and the digitalisation of public administrations & improvement of government transparency.

In **PEACE AND SECURITY**, while the growing climate-induced crises highlight the need to sustain and increase the use of Earth Observation for disaster management and environmental protection, other pressing policy priorities are increasingly addressed by applications based on Earth Observation data, from defence and conflict prevention to post-crisis reconstruction & recovery. Countering disinformation with trusted Earth Observation data also emerges as a priority, in parallel to continuous treaty verification. Other opportunities include using Earth Observation for development & humanitarian crisis cooperation, pandemic and health crisis response, and diplomatic negotiations.

In **CLIMATE RESILIENCE**, a systematic and comprehensive role of Earth Observation is manifested through the significant level and uniform distribution of projects across the identified use cases. This aligns with a broad integration of Earth Observation in climate policies and its ability to structurally address both the causes and effects of climate risks. Despite the wide distribution of solutions, opportunities are observed with regard to reducing species & habitat stress and addressing diseases & pest outbreaks. Water-related issues particularly stand out, from water scarcity to sea water intrusion. In a context of shifting policy priorities, there is also an opportunity to further leverage established applications for civil security purposes and as enablers and main drivers of bilateral and regional development cooperation efforts.

In the **INSURANCE** sector, considerable alignment is observed between risks covered and available Earth Observation applications, signalling a growing opportunity to more systematically leverage Earth Observation to support operational risk reduction, especially in areas within the climate resilience and security domains. Opportunities include severe weather events, coastal and inland flooding, terrain instability, wildfires, protection of critical infrastructure and disaster response.

Against this background, ESPI's next steps include **deep dives** into priority policy areas and sectors, starting from **Water, Energy** and **Diplomacy**. These will go beyond raising awareness and building capacity of policy and industry stakeholders, but also attempting to identify and unlock enablers of the identified opportunities, in terms of public funding streams, commercial market drivers, and public private partnerships.

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1 NAVIGATING A POLICY DOSSIER FOR EARTH OBSERVATION

Space solutions, particularly through Earth Observation, are contributing to the achievement of Europe's priority objectives in most policy areas, from climate adaptation to urban development and from food security to defence. This report, structured as a dossier of opportunities, aims to enhance understanding of the value that Earth Observation solutions bring to European policies, examining how this value extends across different policy domains and contributes to specific objectives within each of them. The research builds on the understanding that scaling the benefits of Earth Observation requires an initial analysis of how derived data and services are currently used in achieving public policy objectives.

This document outlines the outcomes of four exercises:

- The analysis of **policy priorities** of European countries and the identification of specific objectives (*hereafter* 'use cases') within selected domains, based on the screening of more than 300 policies.
- The compilation and review of over 600 **projects based on Earth Observation data** and their main objectives within the selected domains, followed by the comparison of the level of policy prioritisation and number of projects for each use case leading to the identification of potential opportunities.
- The development of five case studies on the practical benefits of Earth Observation data and solutions for public administrations in Europe.
- A complementary assessment of the extent to which Earth Observation is integrated into policies.

> The term '**projects**' herein refers to **ongoing or planned solutions based on Earth Observation data**, including feasibility studies, demonstrators or pilots, programmes, online tools or platforms, while not including upstream capabilities (See further in Annex A – Extended Methodology).

The six domains ultimately covered in this study are listed and briefly described below:

Energy
Infrastructure & Renewables

Including the upgrade and monitoring of energy infrastructure (grids, poles, etc.), the transition to renewable energy (including the phasing out of fossil fuels and the development of clean methods), nuclear energy safety, decommissioning, and safety, energy efficiency, and the security of the energy supply. Also includes the upgrade and monitoring of marine energy infrastructure.

Climate Resilience
and (Natural) Emergency Management

Focuses on human adaptation measures in response to climate changes. Includes climate-induced natural disasters such as floods, wildfires, and drought, and other natural disasters such as earthquakes, tsunamis and hurricanes and their management and disaster planning. Also includes species migration and habitat stress, urban heat island effect (including air quality degradation), water scarcity and degradation, and the spread of diseases.

Agriculture & Food

Including the monitoring and operations management of farming, land, and livestock (including aquaculture), the development of farmers and rural farming, farm sustainability and biodiversity, and the food supply chain from farm to consumer.

Peace & Security

Combines defence, law enforcement & justice, and civil security, including human rights, the migration and protection of citizens, international diplomacy, protection from threats, and the mobilisation and strengthening of defence capabilities. Also includes the preservation of natural resources, natural hazard disaster planning and response, and the mitigation of risks to essential systems (i.e., food, water, and energy security.)

Digital Transition

Including the expansion, monitoring, and upgrade of digital infrastructures and technologies, and ensuring that their supply chains become more sustainable, resilient, and energy- and resource-efficient. Includes digital upskilling, digital coverage for remote areas, and the development of new technologies such as AI and machine learning, quantum, cloud computing, and big data.

Finance & Insurance

For finance, including risk evaluations for loans or lending, asset monitoring or valuations, investment decision making (e.g., urban planning) or space as a watchdog for ESG. For insurance, including fraud prevention, risk assessment and claims verification.

1.1 Key Metrics & Insights

An overview of the policy and project collection activities is shown in Figure 1. A dedicated summary of the main statistics for each domain is provided at the beginning of the following chapters (§§2-7), followed by more detailed policy, project and gap analyses.

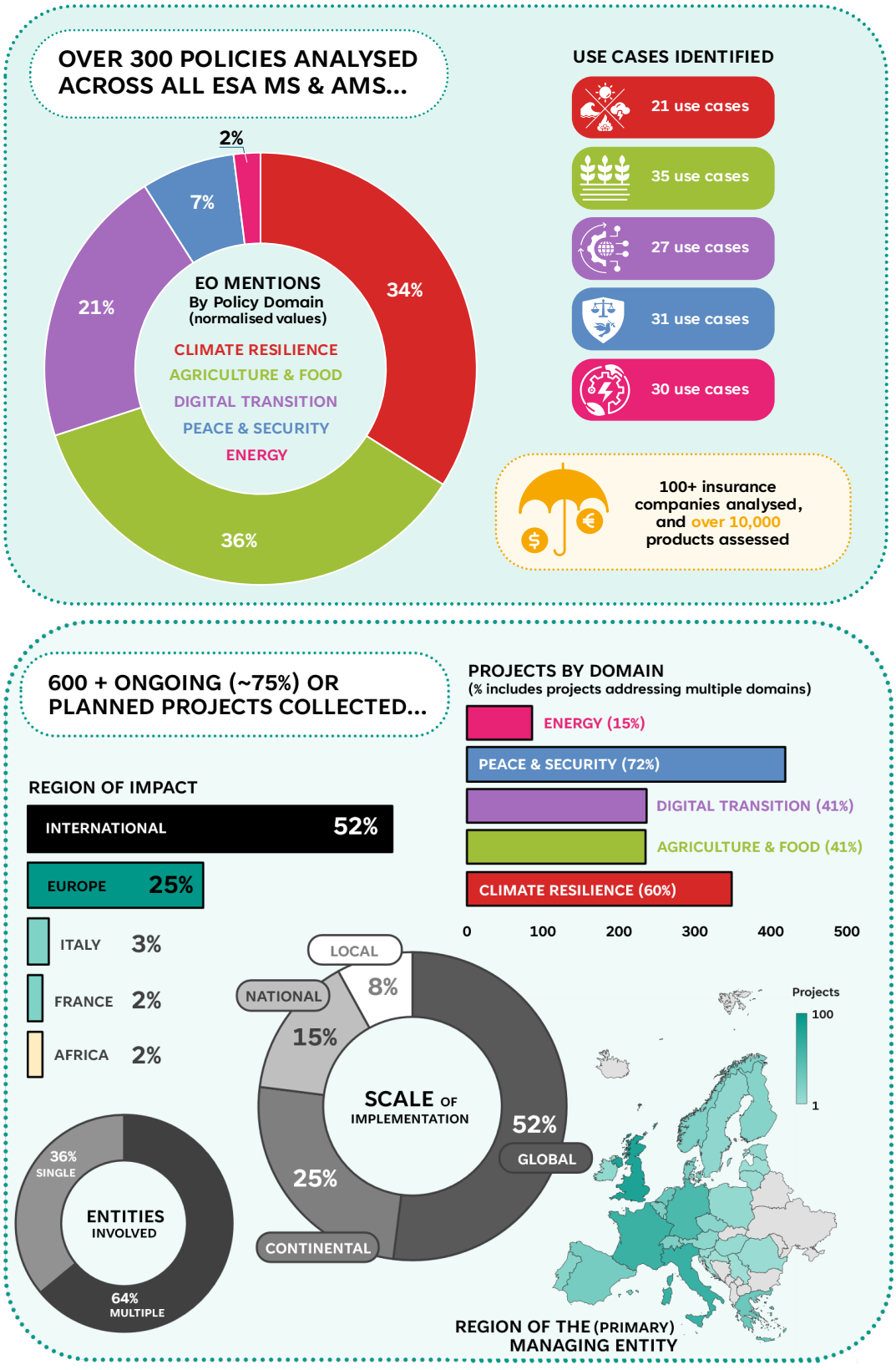


Figure 1: Overview of policy and project collection activity (Definitions in Annex A)

Highlights

- **Climate and Agriculture policies** show the **highest level of integration of Earth Observation** by a considerable margin.
- **The Finance & Insurance domain required a tailored methodology** (See Annex A – Extended Methodology), with the **results of the analysis complementing the study of the other five domains**, more than producing a self-standing outcome (See §7).
- The top domains addressed by the analysed projects are **Climate Resilience** and **Peace & Security**, the latter being heavily influenced by the high number of projects addressing the 'Preservation of Natural Resources' use case.
- **72%** of collected projects address **use cases across more than one domain** (Figure 2).

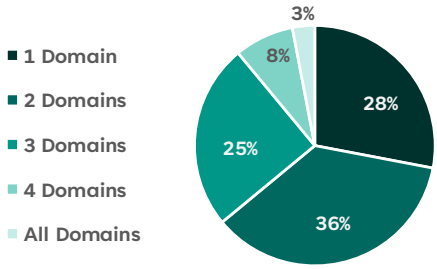


Figure 2: Transversality of projects across policy domains

- The top managing regions across all the collected projects are the **UK** (14%), **Italy** (11%), and **France** (11%).
- The majority of projects have an **impact beyond national and European borders** (Figure 3). Africa stands out across the non-European regions benefitting from collected projects.[2]

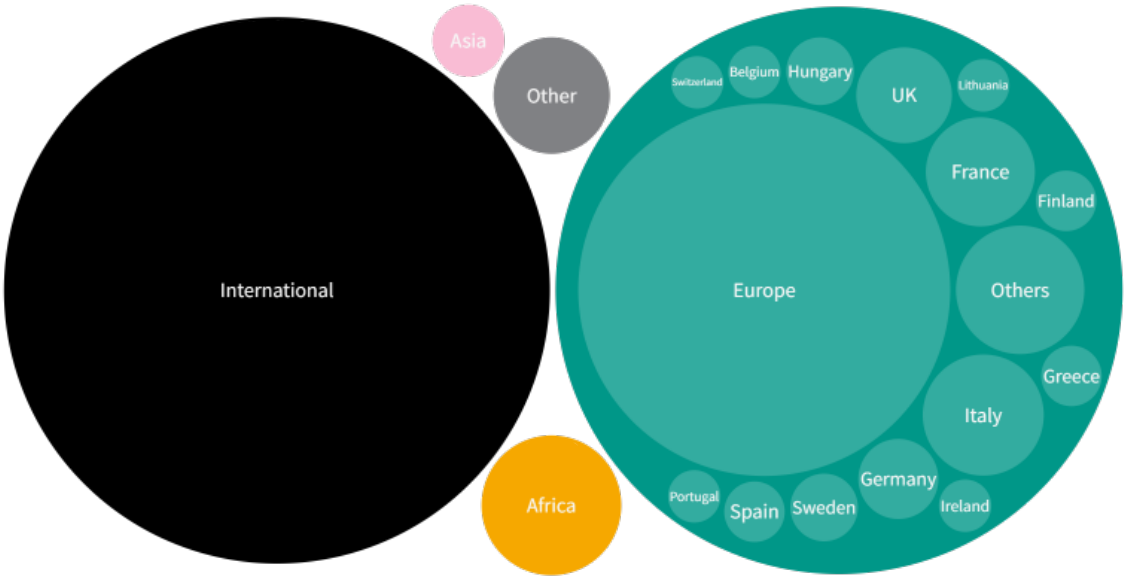


Figure 3: Regions of impact of collected projects (bubble size = number of projects)

[2] International indicates more than one country impacted across different continents; Europe indicates projects impacting more than one country in Europe; Africa/Asia indicates the continent of the impacted country. For further details See Annex A.

1.2 Scope & Methodology

The study was limited to the Member States and Associate Member States of the European Space Agency (ESA), whose experience and feedback were instrumental in refining the results. Six policy domains were examined, relying on general policy documents, i.e. excluding use-case-specific and binding instruments. In terms of space applications, only projects based on Earth Observation were selected, including those integrated with other space (e.g. navigation) or non-space technologies (e.g. drones). Both ongoing (~75%) and planned (<5%) projects were collected and analysed, including a notable share (>20%) for which information on the exact starting date was limited.

The selection and definition of the six policy domains were informed by broad priorities in European policymaking, also influenced by interests and trends in industrial sectors (e.g. Energy). The selection was primarily based on the **assessment of short-term national and European high-level objectives** and complemented by an analysis of global Earth Observation market segments (See further in Annex A – Extended Methodology, including other domains considered in the down-selection but not shortlisted). The domains were defined to minimise overlaps that could introduce unnecessary complexity in the domain-specific analyses. Nevertheless, some overlap of use cases across domains remains (e.g. disaster management objectives appear in both Climate Resilience and Peace & Security), reflecting the inherent links between certain policies.

The policy and project analysis, further described in Figure 4, were primarily conducted through desk research, complemented by more than ten expert interviews to validate the results, fill data availability gaps and provide deeper insights into particularly relevant cases (See Annex A). Insights from the policy workshop at the Royal Society (See Takeaways & Path Forward) have also significantly informed the understanding of the policy landscape and related opportunities.

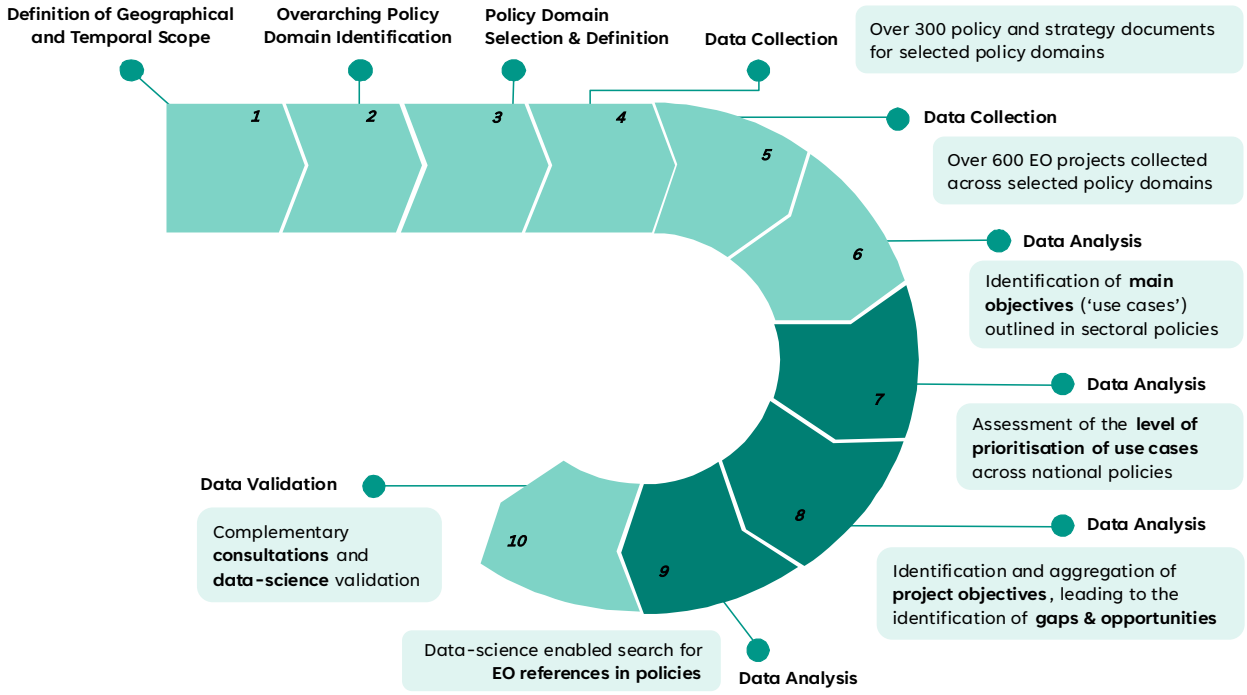


Figure 4: Overview of the methodology steps (main data analysis steps in dark teal)

In this context, the focus was placed on **primarily identifying potential opportunities more than gaps**. This accounts for the fact that some apparent 'gaps' in comparing the level of policy prioritisation and number of projects reflect limited applicability of Earth Observation rather than unmet potential for additional benefits.

In parallel, the policy documents were reviewed using various data science methods to find references to Earth Observation in the context of the implementation of the policy. This allowed a comparison across different domains, as well as across countries within a specific policy area.

Finally, the five case studies to evaluate the practical impact of Earth Observation, and in particular Copernicus, on the activities of public administrations in Europe involved the consultation of selected entities to assess the qualitative and quantitative socio-economic and environmental benefits derived from the Copernicus data (See further in Annex A).

Overall, the outcomes of this research can serve as a **basis for identifying the use cases and domains where new projects based on Earth Observation data could be initiated**, while not providing an assessment of the technical or market feasibility of the solutions potentially filling existing gaps. Moreover, the study can support **policymakers to have a more informed overview of the space applications available to serve their priorities**. The results can also support national authorities in comparing national interests to the gaps and opportunities for project development and space-enabled policy action at the European level.

Nevertheless, this study represents only an initial step in assessing the role of Earth Observation for national policies in Europe. Indeed, while aiming for a comprehensive analysis within the selected domains, the identification of policy objectives and collection of projects face several constraints, in particular related to language or very limited impacts. Moreover, all projects collected are given equal weight when calculating the number of projects addressing a specific use case, regardless of their size, budget, stage of development and timeframe. Further assessment of these parameters would be required to fully assess the policy impact of Earth Observation (See Takeaways & Path Forward). Nevertheless, the list of objectives shared by policy and Earth Observation projects provides a **common language for further advancing research** and engagement with relevant actors.

1.3 Report Overview

The report is structured across dedicated chapters on each of the six selected domains (§§2-7), followed by a second part on the integration of space into sectoral policies (§8). These analyses inform the overarching Statistics and Highlights provided in the previous sections and the Takeaways & Path Forward at the beginning of the document.

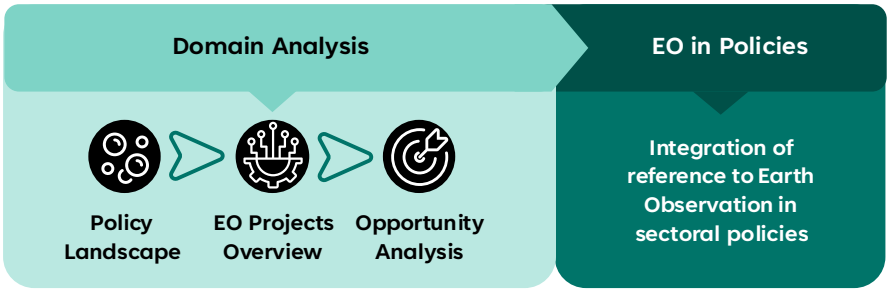
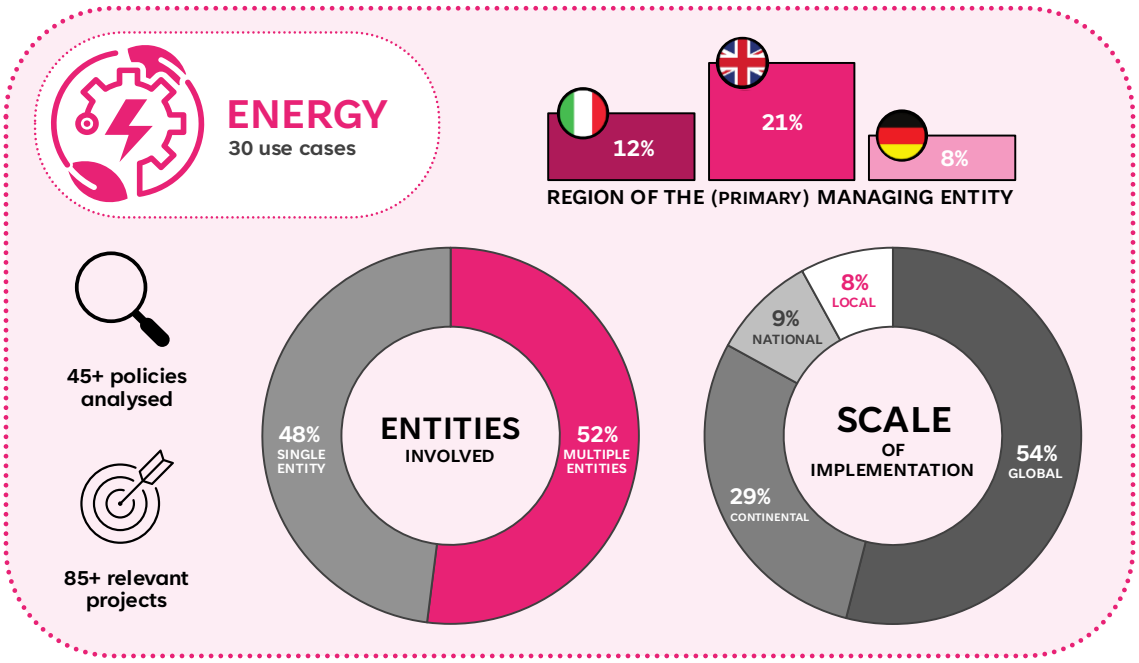


Figure 5: Structure of the main body of the report

The assessment of the contribution of collected projects to each of the selected policy domains is carried out in three main steps. These start from the analysis of the Policy Landscape, conducted prior to assessing any links with Earth Observation, in order to build the foundations for the subsequent space-policy analysis, i.e. Project Overview and Opportunity Analysis. Where relevant, these are followed by the case studies on the benefits of Earth Observation for public administrations.

2 ENERGY



General insights

- **Top policy priorities** are related to **Energy Infrastructure** monitoring, upgrading & modernising, and expansion.
- **Most Earth Observation projects** fall under the use case categories of **Energy Security** and **Renewable Deployment.**
- Limited references to Earth Observation in Energy policies across all analysed countries.
- A very **limited number of projects** are relevant for Energy – only 15% of the total collected. The Energy domain has the **highest share of commercial** projects (>30%), and the **lowest share of academic** projects (~1%).
- **Northern Europe** accounted for the **largest share of projects (33%)** in the Energy domain, with the UK alone accounting for 21%.

2.1 Policy Landscape

Policy priorities in the Energy domain range from objectives related to the management and development of **energy infrastructure** to the development and deployment of **renewables**. Use cases linked to increasing energy **efficiency**, environmental **sustainability** and **security** of the energy system also appear as priorities in the energy policies of European countries (See the objectives identified and prioritised by the analysed countries in Figure 6).

Overall, there is a strong prioritisation of **renewables deployment and energy infrastructure modernisation**. Grid flexibility, smart networks, and storage solutions are viewed as essential for integrating intermittent renewable sources. In contrast, nuclear energy receives significantly less emphasis.

In the context of environmental sustainability of energy systems, there is an emphasis on the low-carbon transition and the modernisation of energy infrastructure, with use cases including **energy efficiency in buildings and industry**, **electrification of transport**, in addition to renewables and grid modernisation. In this regard, priority is also placed on **phasing out fossil fuels rather than optimising their use**; indeed, policies tend to focus on replacing fossil fuel consumption through electrification, modernisation and renewables, instead of cleaner technologies still based on fossil fuels. This aims to reduce both emissions and import dependency, thus also aligning with the broader **priority of energy security**.

Moreover, the identified national priorities also align closely with EU-level strategies such as the EU Green Deal, the REPowerEU plan, and the 2030 climate and energy framework, which collectively aim to advance energy independence, emissions reduction, decarbonisation, and energy system resilience.[3] This alignment partly reflects the use of National Energy and Climate Plans, where available, for the identification of the priorities of the analysed countries.

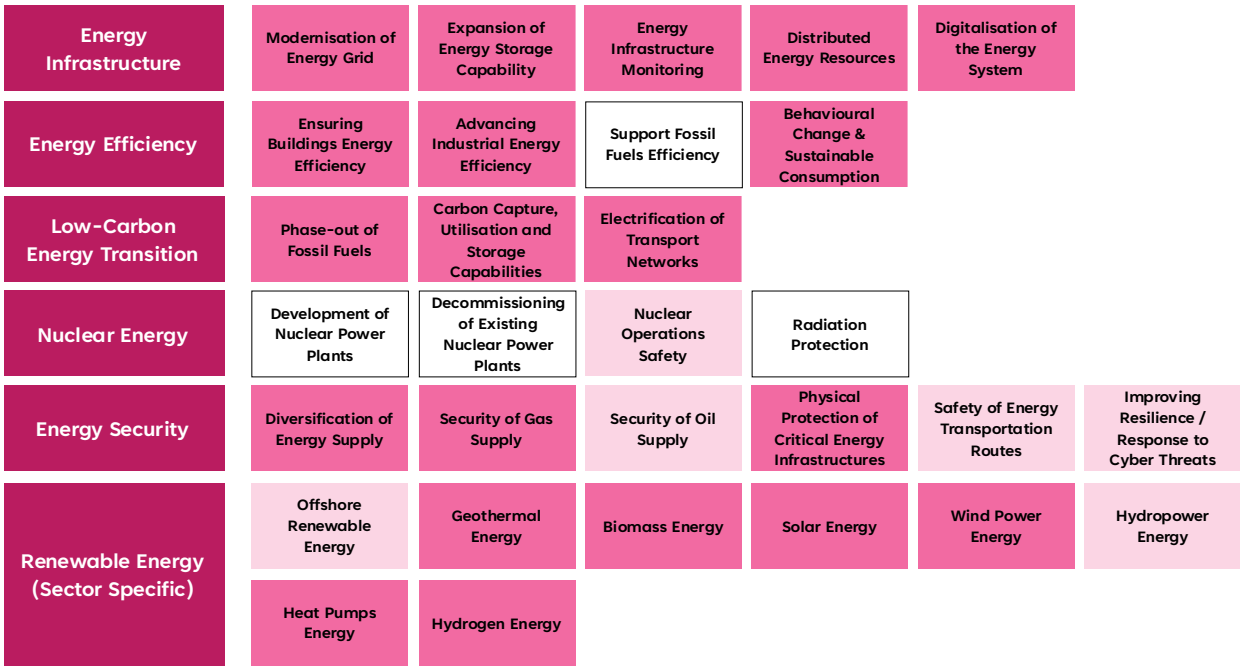
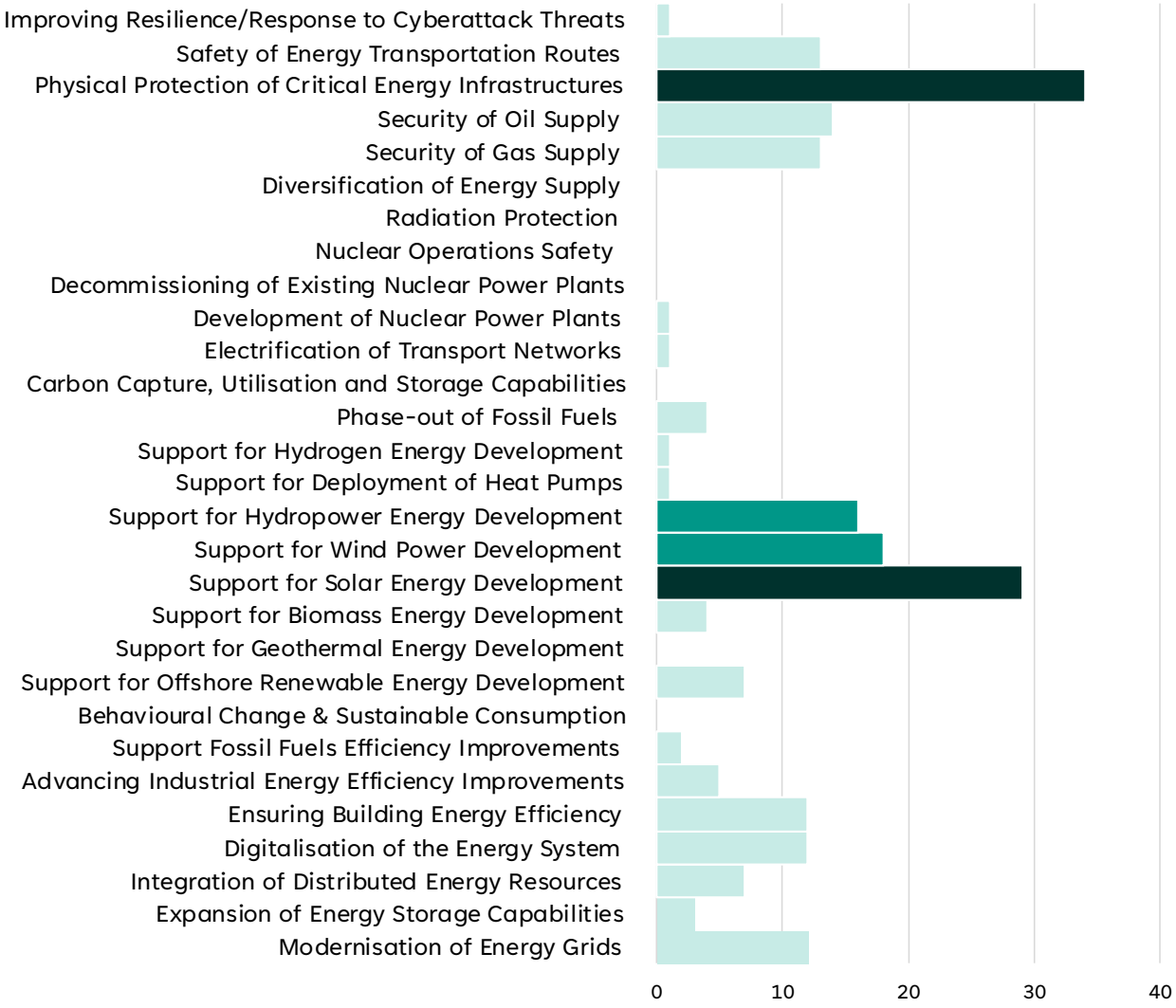


Figure 6: Energy use cases prioritised by >65% (light pink) and >90% (pink) of analysed countries across the six categories outlined on the left (dark pink)

[3] 'The European Green Deal' - European Commission; 'The 2030 climate and energy framework' - Consilium

2.2 Earth Observation Projects Overview

The Earth Observation projects directly relevant to the Energy domain (around 100 identified) are distributed across use cases as shown in Figure 7. The *Energy Infrastructure Monitoring* use case is shown separately as an 'outlier', since it is addressed by more than half of all identified projects, reflecting its particularly broad scope. The distribution below only includes projects explicitly mentioning one or more of these use cases within their objectives – it does not account for any indirect or potential relevance of projects for other use cases.



Outlier: *Energy Infrastructure Monitoring*, with more than half of the Energy projects (>45 units)

Figure 7: Distribution of Earth Observation projects across Energy use cases, categorised using a three-tier scale indicating the number of projects per use case

Overall, projects are mostly concentrated in two categories of use cases: (1) **infrastructure monitoring and security**, with a focus on the protection of critical systems, and (2) the **deployment of renewables**, particularly for solar, wind, and hydropower, but also with some contribution to biomass and offshore plants.

While these remain high-priority use cases, the comparatively limited contribution of Earth Observation projects to other key objectives (See specific gaps in §2.3) highlights the **need for a**

more systematic and comprehensive contribution of Earth Observation solutions to energy policies. Increasing understanding of the potential value of Earth Observation among policymakers in the Energy domain would also contribute to strengthening alignment between national priorities and available Earth Observation solutions.

A substantial share of the collected projects are relevant to more than one priority use case **within the Energy domain**, indicating an **expanding scope of monitoring capabilities beyond operational and efficiency performance**. For instance, several projects contributing to energy infrastructure monitoring also support security objectives, particularly the physical protection of critical energy systems, the security of oil and gas supply and energy transportation safety. Similarly, infrastructure monitoring solutions also contribute to renewables deployment (especially solar and wind), showing a tendency to integrate monitoring with broader energy system planning.

Moreover, a high number of Earth Observation projects already contributing to **other domains**, especially Peace & Security, also cover Energy use cases. For instance, a significant share of projects focused on protecting and strengthening critical infrastructure place specific emphasis on energy systems. Similarly, projects supporting natural hazard response (incl. droughts, floodings, urban heat islands effects), relevant for both Peace & Security and Climate Resilience, also address objectives of energy efficiency, systems resilience and smart deployment. An example is provided by thermal imaging and surface temperature mapping to identify urban heat island hotspots, as performed by several reviewed projects. Furthermore, many projects contributing to real-time assessment and decision-making objectives (Digital Transition) are specifically focused on energy infrastructure monitoring and protection. Projects using Earth Observation for predictive maintenance, extreme weather risk detection, or threat identification, demonstrate the added value of combining satellite data with digital solutions in this context.

Overall, these findings highlight a potential for the **design of Earth Observation projects targeting specific Energy-related use cases with even broader cross-domain policy appetite and impact**. Moreover, they illustrate the versatility of Earth Observation technology and solutions, which can **serve even unrelated objectives with tailored applications derived from the same instruments and datasets**, such as in the case of projects relevant for both unconnected Energy and Agriculture use cases, enabling more efficient use of resources. Realising this cross-domain potential requires a broad understanding of policy needs as well as of the possible uses of available Earth Observation data.

2.3 Opportunity Analysis

The comparison of policy priorities with the ongoing and planned projects addressing them (See §§2.1 & 2.2) enables the identification of gaps in the contribution of Earth Observation solutions to Energy policies, representing areas where current project implementation does not yet align with established priorities and related opportunities. The assessment of the technical feasibility of solutions filling existing gaps is beyond the scope of this study.



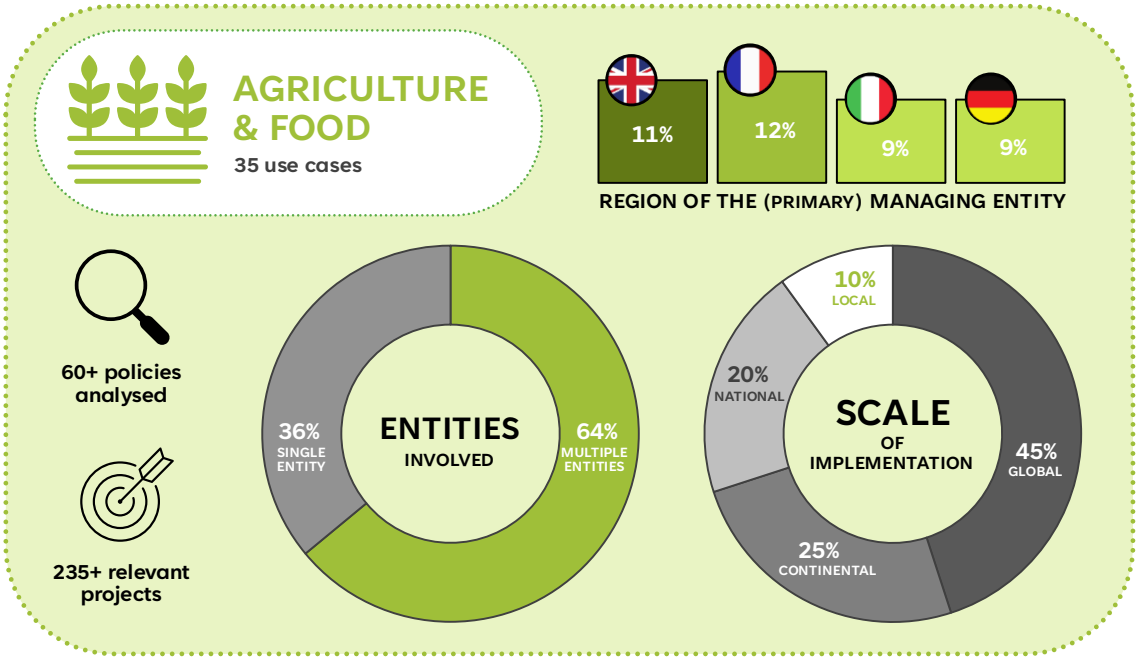
Overall, the energy domain shows relatively **low alignment between policy ambition and the contribution of Earth Observation projects to their attainment**, especially when compared to other domains such as Climate Resilience & Emergency Management or Agriculture & Food. Indeed, despite the high prioritisation of the Energy domain in public policies, only a limited number of Earth Observation projects address these priorities, resulting in several notable gaps. The **clearest potential opportunities** are visible in:

- *Electrification of transport networks* – Prioritised in national and EU policies, especially under Fit for 55 and REPowerEU, but minimal coverage of projects. Available solutions show opportunities for increasing Earth Observation support to grid connectivity planning, mapping traffic flows and urban density, and informing optimal siting of charging infrastructure.
- *Securitisation of Oil and Gas Supply* - Highly prioritised at national level, particularly in the current geopolitical context, but limited uptake of Earth Observation. Earth Observation has been leveraged for monitoring shipping routes, ports, pipelines, storage sites, as shown by the projects reviewed, and can be used to identify risks to oil and gas infrastructure.
- *Ensuring Energy Efficiency of Buildings and Industrial processes* - Limited contribution of Earth Observation solutions in both these areas is observed, despite the significant potential uses of Earth Observation, also aligned with national and EU policies. For instance, thermal infrared mapping of heat loss is used by municipalities to orient investments in insulation, retrofitting, and heating system upgrades.
- *Diversification of (Energy) Supply* – Significantly prioritised in policies, also as part of the Peace & Security domain, with very few projects explicitly addressing it as their objective. While Earth Observation is already being leveraged directly or indirectly for such purposes (e.g. monitoring maritime transport routes, tracking pipeline and shipping vulnerabilities, and assessing environmental risks), more solutions could be explicitly tailored to ensuring *Safety of Energy Transportation Routes*.
- *Deployment of Renewables* – While Earth Observation solutions are already contributing to this area by supporting the deployment of hydro, solar and wind power infrastructure, projects can be further developed for other renewables such as *Biomass* and *Geothermal* energy. This is also relevant to address the broader policy priority of *phasing-out fossil fuels*.
- *Digitalisation of the Energy System* – Although this broad use case was identified as a top policy priority, only a limited number of projects explicitly address it. Nevertheless, the collected projects show that Earth Observation solutions are already contributing to digital twins, smart grid management, and predictive maintenance. There is significant potential to scale and replicate these successful examples, including in support of more specific objectives such as the *Integration of Distributed Energy Sources* and *Modernisation of Grids*.

Finally, a notable gap emerged in the contribution of Earth Observation solutions to *Carbon Capture, Utilisation, and Storage*. While the contribution of Earth Observation solutions to climate mitigation strategies is substantial,[4] the analysis highlights the lack of an explicit link between these existing efforts and other priority policy domains and sectors, such as Energy, which would increase the perceived benefits of Earth Observation.

[4] This point is not investigated further in this document, whose focus is deliberately shifted to Climate Resilience, instead of Mitigation, to identify additional opportunities for Earth Observation beyond the established contribution to mitigation actions.

3 AGRICULTURE & FOOD



General insights

- **Top policy priorities** are distributed across a wide range of areas, with **Farming Development and Support** emerging as key policy priorities.
- **Most Earth Observation projects** fall under the categories **Land & Crops** and **Farm Monitoring & Management**. A **substantial share is led by Western European countries** (>40%).
- **High level of integration of Earth Observation into Agriculture & Food policies**, alongside Climate Resilience.
- **Highest percentage of academic** initiatives across domains (~5%) coupled with **highest percentage of civil institutional** ones (28%), similarly to the case of Climate Resilience.

3.1 Policy Landscape

The main areas addressed by the Agriculture & Food policies cover (1) the **management of farming activities**, including effective monitoring, forecasting, development and resourcing thereof, (2) the assessment of the **quality of resources (soil, water) and products** derived from agriculture, (3) the **environmental sustainability** of such activities, and (4) the **safety and resilience of the food value chain** (See the objectives identified and prioritised by the analysed countries in Figure 8).

Policy priorities in the agricultural domain show a general tendency to focus on the **modernisation** of the sector and **environmental preservation**, opening opportunities for innovation while generating potential tensions between these two overarching objectives. Indeed, scaling organic, local, and sustainable methods may require significant resources and the integration of advanced technologies in agricultural practices, with the risk of leaving small-scale farms behind – especially concerning the access to the new tools and the specific skills required for their exploitation. In this regard, policies prioritise the **promotion of new generations**

of farmers and their training, which should be supported by technological tools enabling their integration into operational agriculture practices.

The emphasis on **sustainability and ecological health** is reflected in several prioritised use cases (i.e. ensuring soil and water body quality, biodiversity protection, and promoting organic, local and seasonal produce), which call for new and innovative farming methods with lower environmental impact. This also represents a policy focus on strengthening **long-term resilience of the agricultural systems and the food value chain**, closely aligned with climate resilience priorities (e.g. water scarcity and quality degradation, terrestrial ecosystem degradation, and soil degradation and desertification). Similarly, agriculture-related monitoring, forecasting, and early-warning systems can reduce reliance on pesticides and thereby contribute to reducing the spread of diseases and pest outbreaks, thus benefitting from broader policy appetite also within the climate resilience domain. Therefore, tools contributing to broader ecosystem resilience are strategically positioned at the **intersection of the agriculture and climate resilience domains**.

In this context, there are potential **tensions between rural development and sustainability** objectives, which could benefit from innovative solutions. For example, the demonstrated capacity of Earth Observation to guide infrastructure planning and monitor environmental impacts of agricultural practices could support the design of mutually reinforcing measures pursuing development and sustainability goals.

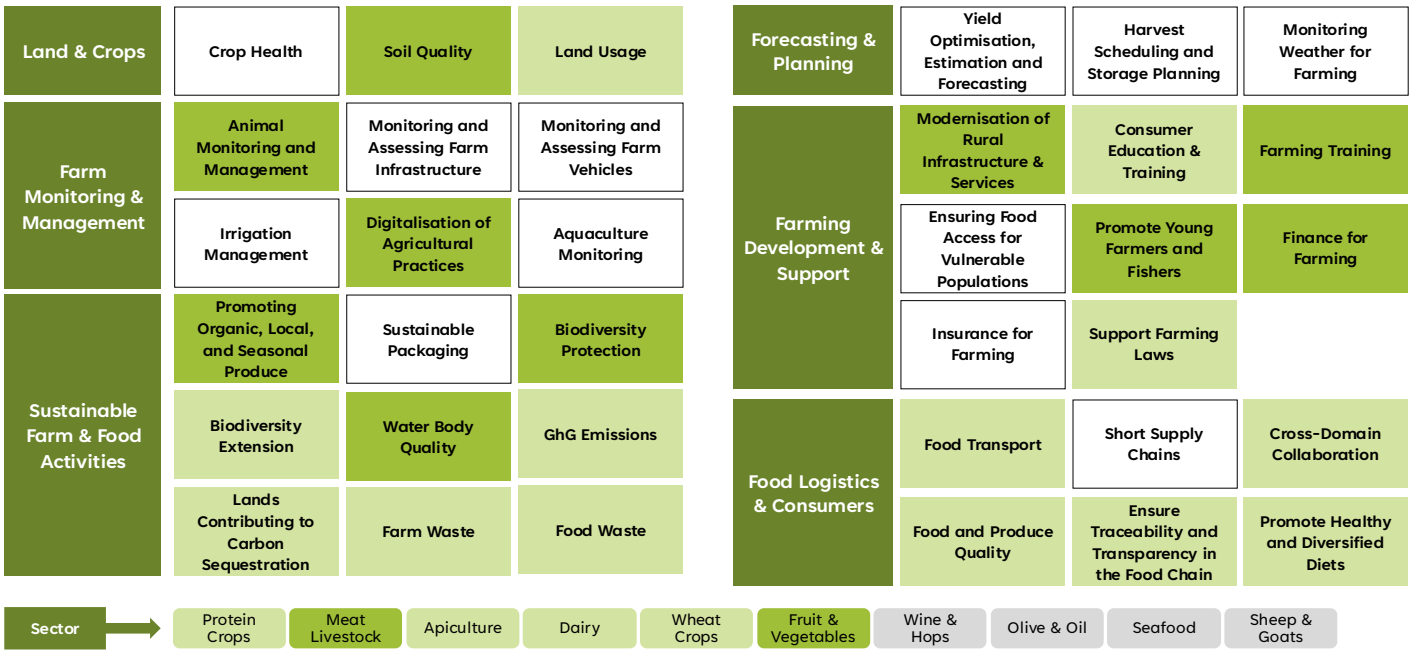


Figure 8: Agriculture & Food use cases prioritised by >65% (light green) and >90% (green) of analysed countries across the six categories outlined on the left (dark green)

3.2 Earth Observation Projects Overview

The Earth Observation projects directly relevant to the Agriculture & Food domain (just under 250) are distributed across use cases as shown in Figure 9. The distribution below only includes projects explicitly mentioning one or more of these use cases within their objectives – it does not account for any indirect or potential relevance of projects for other use cases.

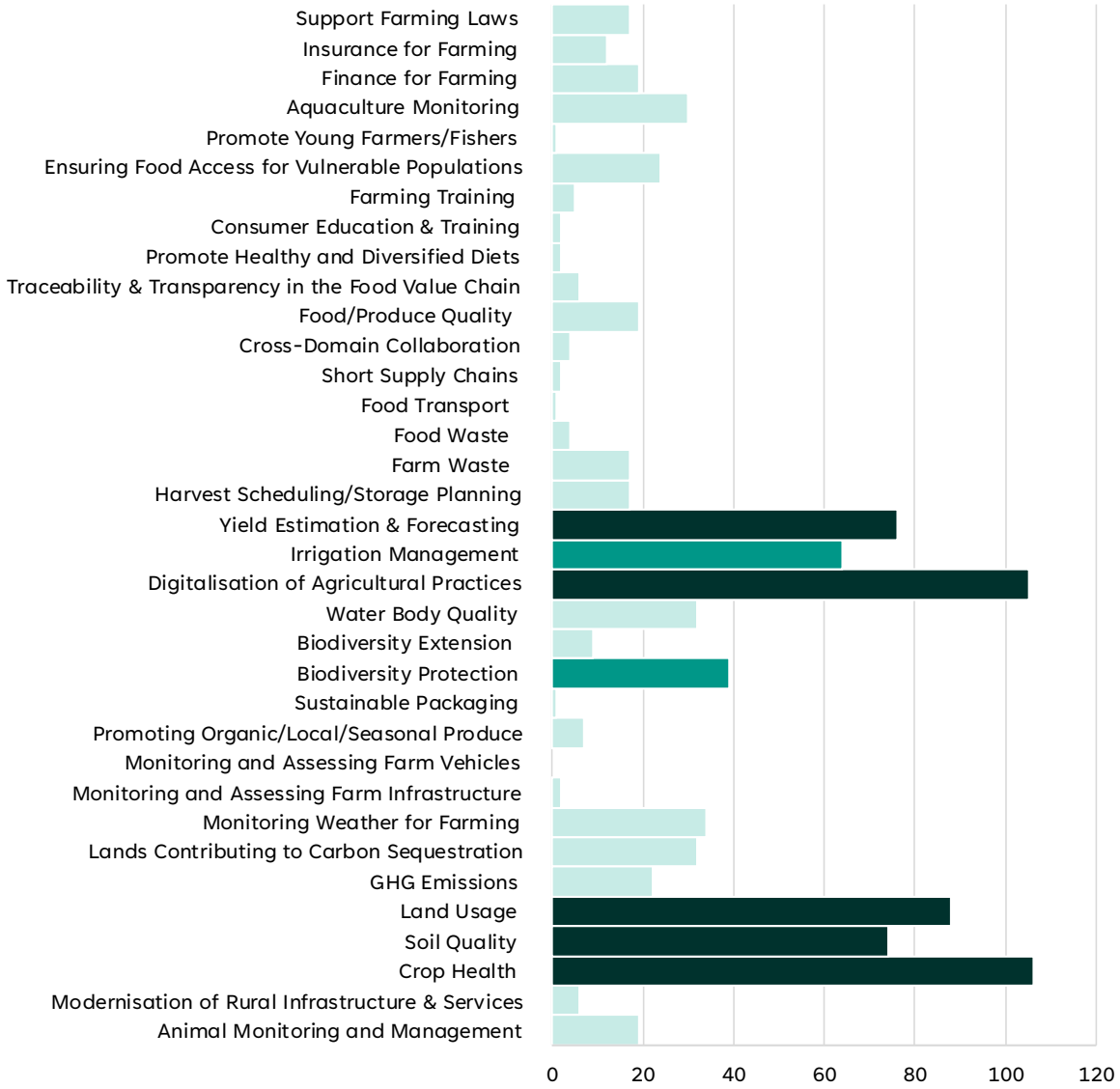


Figure 9: Distribution of EO projects across Agriculture & Food use cases, categorised using a three-tier scale indicating the number of projects per use case

Overall, projects are largely concentrated in a small number of use cases related to agricultural resources (i.e. *Soil Quality*, *Land Usage*) and products (i.e. *Crop Health*, *Yield Estimation & Forecasting*). When implementing digital tools in support of agricultural activities, these projects also contribute to the broad objective of increasing the *Digitalisation of Agricultural Practices*, thus explaining the high number of items relevant for this use case.

In particular, a first insight concerns the significant number of projects consisting of new digital applications based on Earth Observation data and relevant to use cases for productive, resilient, and sustainable farming. Strengthening the alignment between the digitalisation of the

agriculture sector and productive farming practices, especially **at the level of small-scale farming activities**, would help **maximise the impact** of Earth Observation solutions and derived broad data platforms. In order for this approach to be economically sustainable, operational solutions leveraging data platforms should be **modular and replicable**, thus capable of serving similar local needs, with the tailoring process co-designed with farmer associations or other local realities (See also in §3.3). Second, embedding Earth Observation solutions within digital tools for agriculture also contributes, although still to a limited extent, to **supporting farming laws and attracting investments**, with several projects supporting compliance monitoring and subsidy allocation. In this context, some projects also bridge monitoring and support to finance for farming with sustainability objectives, for instance by incentivising sustainable resource management or ensuring compliance with biodiversity protection objectives. However, these links remain partial and open opportunities to be further reinforced.

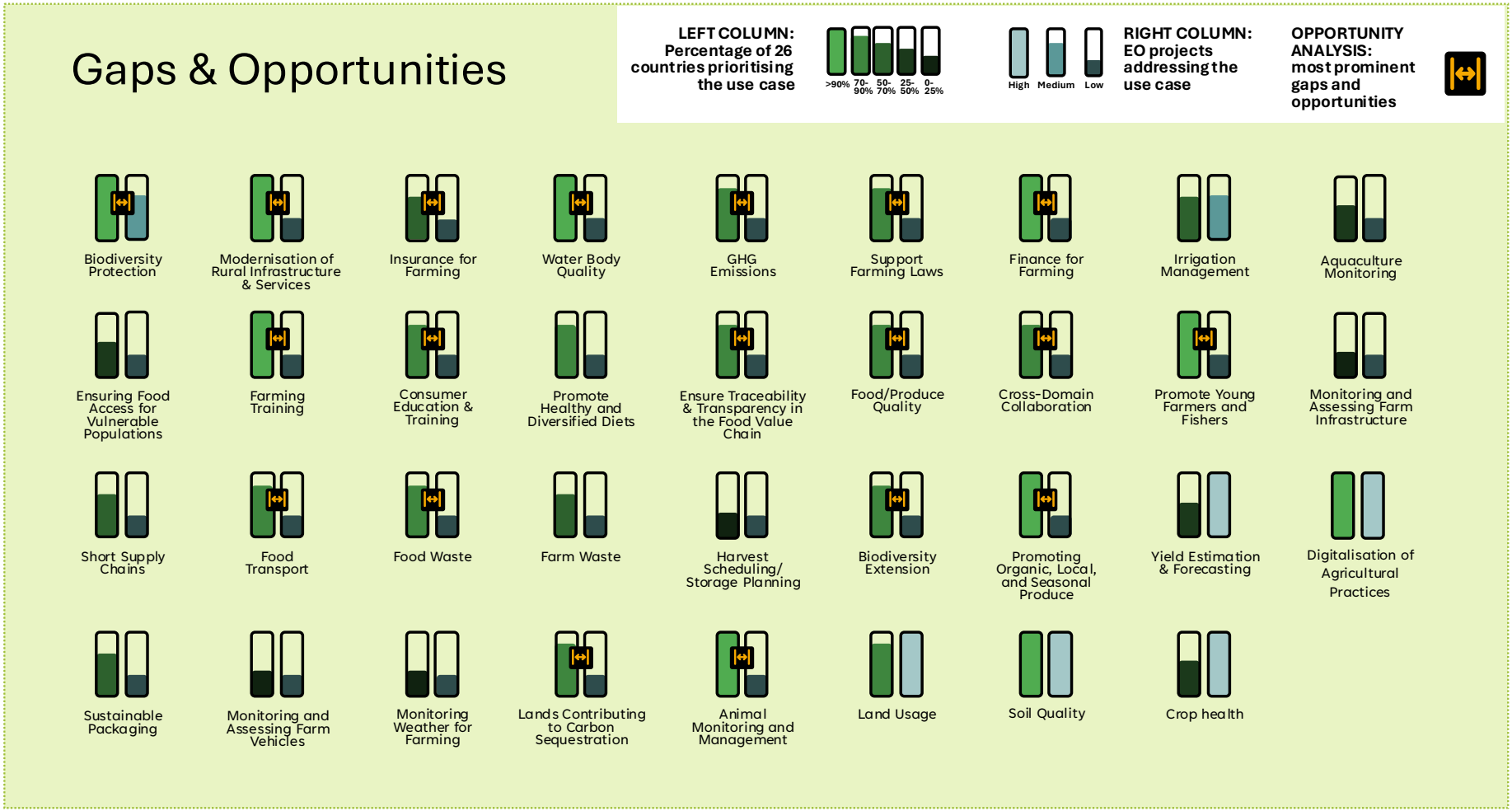
Considering the **cross-domain relevance of projects**, the strong overlaps between the Agriculture & Food and Peace & Security domains reflect links between natural resource protection, natural hazard vulnerability, and food security, in line with the recognition of the food value chain as an essential system.[5] In this regard, there is room for deepening integration, i.e. supporting **Earth Observation projects that simultaneously contribute to both agriculture and security priorities**. Examples emerging from the reviewed projects include using Earth Observation for identifying illegal resource exploitation tied to unsustainable farming practices or, more specifically, early detection of harmful algal blooms for coastal fisheries.

By contrast, relevance of projects across both the Energy and Agriculture & Food domains appears more limited. Frequently this is due to **broad and transversal data platforms covering a wide range of domains with very limited operational impact** in each of them. While opportunities for direct integration are modest, Earth Observation is beginning to provide value in niche areas, such as the assessment of land-use trade-offs between farming and renewable installations or supporting agrivoltaics systems. In parallel, the strong impact of climate stressors on agricultural productivity explains the relevance of a significant number of projects across both the Climate Resilience and Agriculture & Food domains, particularly in relation to drought and water scarcity, and crop yield and resource management. Finally, projects relevant for agrifood also contribute to the achievement of objectives within the Digital Transition domain, such as the uptake and impact of AI in other sectors. Indeed, **Earth Observation data enables and is combined with AI-based and other decision-making tools** to evolve beyond monitoring towards prediction, optimisation, and real-time support (e.g. for precision farming, early warning and yield forecasting). Strengthening integration of Earth Observation with such applications can further support the resilience and sustainability of the agri-food systems.

[5] See its prioritisation as part of the G7 topics in 2024 – G7 Italy

3.3 Opportunity Analysis

The comparison of policy priorities with the ongoing and planned projects addressing them (See §§3.1 & 3.2) enables the identification of gaps in the contribution of Earth Observation solutions to Agriculture and Food policies, representing areas where current project implementation does not yet align with established priorities and related opportunities. The assessment of the technical feasibility of solutions filling existing gaps is beyond the scope of this study.



Overall, there is a substantial baseline of projects addressing many of the Agriculture & Food objectives. However, alignment between policy priorities and ongoing and planned projects is uneven, partly due to significant concentration peaks in several use cases (See further in §3.2). This indicates the **need to boost project development in under-addressed use cases**, particularly where policy ambition is high (and growing). With the current number, scale and scope of projects, greater impact can only be achieved by **translating priority objectives into concrete funding for Earth Observation** through domain-related budgets (e.g. Ministries of Agriculture) and clear implementation guidance. In this context, the concentration of Earth Observation projects in areas such as soil quality, land use, and crop health reflects a traditional role of Earth Observation in agricultural monitoring through biophysical applications, which continue to be essential. Nevertheless, the gaps identified point to the **opportunity to expand the contribution of Earth Observation** solutions beyond these established areas. For example, Earth Observation could be further leveraged to support farming subsidy allocation, facilitate certification and monitoring of organic production, and be embedded into digital services encouraging their uptake among new generations of farmers, as demonstrated by several existing initiatives. Such developments would position **Earth Observation as an enabling tool in the transformation of agricultural systems**, while potentially requiring further technical advancements, especially in temporal and spatial resolution.[6]

Several **specific opportunities** can be identified as follows:

- *Water Body Quality* – High priority in policies, low level of dedicated projects. This contrasts with the proven ability of Earth Observation to monitor parameters such as algal blooms, nutrient runoff, sedimentation, and pesticide pollution, as well as with the availability of Earth Observation solutions for early warnings of water stress, records of long-term changes in freshwater ecosystems, and assessments of land-use impacts on rivers, lakes, and coastal zones. In particular, these capabilities are already demonstrated by **various projects addressing water and freshwater scarcity and quality within the Climate Resilience domain**, where there is generally solid alignment between water-related policy objectives and the use of Earth Observation. To increase their contribution to the Agriculture & Food domain, these projects would need to be more specifically tailored to address agricultural (human) pressures on water systems. This would not only **shift project focus towards protecting essential systems**, with such Earth Observation projects consequently gathering increasing attention in today's policy context, but also support farmers in demonstrating compliance with environmental objectives.
- *Modernisation of Rural Infrastructure & Services* – While also relevant for other space applications (especially telecommunications), the contribution of Earth Observation projects could be further increased in this area, which is also relevant for development cooperation efforts. Current projects leverage Earth Observation for rural infrastructure investments planning and monitoring, environmental impact assessment, and ecosystem services assessment.
- *Food Value Chain* – Beyond agricultural practices, Earth Observation projects show slower progress in contributing to several use cases across the food value chain, which are currently prioritised in food security policies and appear among diplomacy and international relations

[6] 'Evolving Earth observation capabilities for recent land-related EU policies' - ScienceDirect

objectives. This requires further development of Earth Observation solutions supporting objective in areas such as food transport, traceability and waste management.

- *Promotion of Organic/Local/Seasonal Produce* – This use case is prioritised by all analysed countries, but coverage by Earth Observation projects remains limited, partly due to challenges associated with contributing effectively at the local scale. Existing projects with potential in this area provide guidance to farmers on optimising fertiliser use to reduce synthetic nitrogen inputs, support local authorities in certifying organic plots, and contribute to enhancing supply chain transparency and strengthening consumer trust.
- *Biodiversity Protection and Extension* – This objective is prioritised across both the Agriculture & Food and Climate Resilience & Emergency Management domains. However, the number of Earth Observation projects addressing Biodiversity within the specific context of the agriculture domain is more limited, again highlighting a need to shift the focus from solely biodiversity protection as part of climate action to **protecting and increasing the benefits of biodiversity for (the resilience of) essential systems**, such as water and food.
- *Animal Monitoring and Management* - Relatively low alignment between policy focus and Earth Observation project coverage, despite the existence of precision livestock farming technologies, **also enabled by other space applications such as GNSS** (e.g. tracking collars and smart feeding systems). As demonstrated by several projects, Earth Observation could be further integrated for monitoring pasture quality, providing early warning, and predicting food and water shortages, especially when complemented by *in situ* data and sensors.

More broadly, a key requirement for Earth Observation projects to better contribute to the achievement of the above-mentioned priorities concerns the **development of solutions capable of being implemented at the local level while designed to be scalable by replicability** in other contexts. This differs from local pilot initiatives without clear path for expansion, but also contrasts with only relying on broad data repositories with wide scope and geographical coverage but limited capacity to enable services supporting specific realities and needs without more operational solutions embedded, as confirmed by expert interviews.[7] Such an approach would also facilitate the **integration of *in situ* data**, which is key for tailoring solutions and thus fostering uptake. In this context, leveraging Earth Observation to **fill the gaps in the promotion of new generations of farmers and fishers, as well as providing them with advanced training,** should form part of the pathway for bringing space closer to farmers. This can be achieved by increasingly integrating Earth Observation into agricultural education, embedding it within user-friendly platforms tailored to new entrants, and using information derived from Earth Observation as evidence in financing schemes for young farmers.

[7] This aspect was also investigated as part of additional work conducted by ESPI in 2024 on the topic of Space for Climate Resilience and Food Security in the Mediterranean region.



UPTAKE IN PRACTICE – SLOVENIAN NATIONAL PAYING AGENCY FOR AGRICULTURAL MARKETS AND RURAL DEVELOPMENT

About the Slovenian National Paying Agency

The Slovenian National Paying Agency for Agricultural Markets and Rural Development (ARSKTRP) operates under the Slovenian Ministry of Agriculture, Forestry, and Food.

The **responsibilities** of ARSKTRP include ensuring compliance with the Common Agricultural Policy (CAP), by operating the Area Monitoring System (AMS) to oversee agricultural parcels, verifying land use, and maintaining and updating the Land Parcel Identification System (LPIS). Additionally, it manages the national Farm Registry Service, which collects and maintains up-to-date information on all farms and agricultural holdings in the country.

Data sources – To perform its activities, ARSKTRP relies on Copernicus data, accessed through national or European platforms**, and paid Planet data**.

Benefits of Copernicus data and solutions

User Maturity Level 5 – Operational User [8]

Several examples of how Earth Observation solutions specifically based on Copernicus data are supporting the operational activities of ARSKTRP are provided below, complemented by needs for improvement:[9]

- **Significant efficiency improvement** **in farm controls and subsidy allocation.**[10]
- **Slight improvement** **in handling non-monitorable aspects in farm controls.**[11]
- **4-6 hours saved** **per average field inspection** compared to traditional methods alone.

Outstanding needs for future improvements

- Improved integration of Copernicus with national data sources for data harmonisation and improved detection of changes
- Enhanced Copernicus spatial resolution and improved processing algorithms for the identification of parcel boundaries, and the identification and assessment of small/fragmented parcels (<0.5ha)
- Targeted capacity-building to enhance digital literacy and service uptake in support of the ageing farmer population
- Reduced initial costs and administrative burdens of AMS implementation to increase accessibility

[8] Full integration of Copernicus-based solutions into their standard processes, where the necessary resources, including staff, budget, and facilities, are already allocated or readily deployable, as assessed by ARSKTRP.
[9] The assessment is based on consultations with ARSKTRP and subsequent validation by ARSKTRP of the information hereby provided.
[10] ARSKTRP assessed this improvement scoring it 3 on a scale of 1 (minimum) to 4 (maximum), while highlighting the high cost of setting up the space-based monitoring mechanism, with limited benefit for small countries.
[11] ARSKTRP assessed this improvement scoring it 2 on a scale of 1 (minimum) to 4 (maximum). 'Non-monitorable aspects' include small plots, areas with significant tree coverage, grazed pastures, and moist and soggy soil during slow-growth periods.



UPTAKE IN PRACTICE –
ITALIAN AGRICULTURAL PAYMENTS AGENCY

About the Italian Agricultural Payments Agency

The Italian Agricultural Payments Agency (AGEA) is a government body under the Ministry of Agriculture, Food Sovereignty and Forests, with approximately 500 employees.

The **responsibilities** of AGEA include the management and distribution of EU funds for agriculture, in parallel to ensuring compliance with EU regulations. Therefore, AGEA serves as both a Coordination and Paying Body, in particular supporting the implementation of the Common Agricultural Policy (CAP) by monitoring field parcels and verifying aid eligibility.

Data sources – To perform its activities, AGEA primarily relies on the Area Monitoring System (AMS) based on radar and optical imagery provided by Copernicus Sentinel-1 and Sentinel-2 and accessed via the Copernicus Data Space Ecosystem and national platforms. AGEA was the first EU body to introduce satellite monitoring for Basic Payment Scheme (BPS) controls in 2018.

Benefits of Copernicus data and solutions **User Maturity Level 5 – Operational User** [12]

Several examples of how Earth Observation solutions, specifically based on Copernicus data, are supporting the operational activities of AGEA are provided below, complemented by needs for improvement:[13]

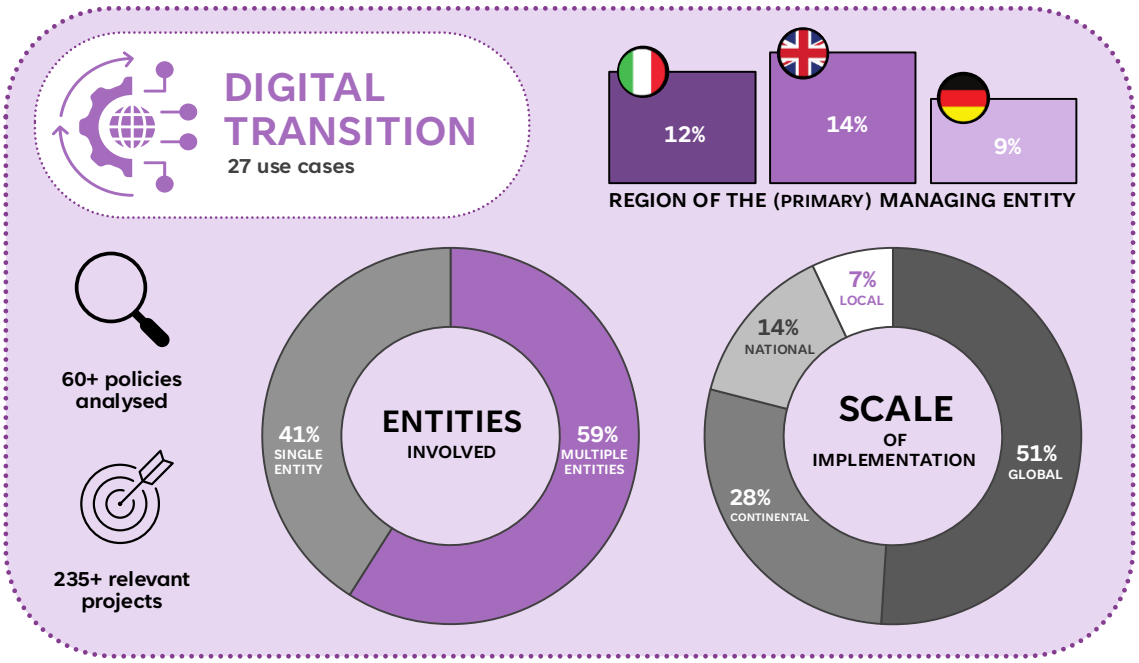
- **More than 90% of land monitored with reduction to 3-5% in-situ checks for direct disbursement**, and gradual replacement of verification activities for rural development.
- **More than 2 days saved per average field inspection** for funds disbursements and **overall 30-40% cost reduction** for monitoring and inspecting activities.
- **~5 months saved in administrative processing** of farmers' requests.
- **~40% increased accuracy & reliability in land parcel assessment** integrating AI.
- **Frequent positive feedback from farmers** on Earth Observation to reduce burden, cost and time in the exchange with AGEA, with higher confidentiality (e.g. reduced document submission).

Outstanding needs for future improvements

- Higher frequency of revisit to increase timeliness of agricultural monitoring
- Reduced initial costs and administrative burden of Area Monitoring System (AMS) implementation to increase accessibility and timely communication with farmers concerning alerts and evidence submission
- Territorial scalability and integration of data from different sources, enabling uniform monitoring throughout the country. This should be coupled with capacity-building on how to leverage platforms that combine Copernicus data, field sensors, and admin procedures

[12] Full integration of Copernicus-based solutions into their standard processes, where the necessary resources, including staff, budget, and facilities, are already allocated or readily deployable, as assessed by AGEA.
[13] The assessment is based on consultations with AGEA and subsequent validation by AGEA of the information hereby provided.

4 DIGITAL TRANSITION



General Insights

- **Top policy priorities** concern the **Digitalisation of Public Administration** and enabling the **Digitalisation of Agricultural Practices.**
- **Top Earth Observation projects** fall under the categories of Digital Infrastructure, and **Data Usage & Analysis.**
- The **highest number of Earth Observation references** in Digital policies (>60% together) were attributed to **Greece & Latvia.**
- **70%** of projects consist of online **tools or platforms.**

4.1 Policy Landscape

The priorities outlined by Digital Transition-related policies in European countries range from addressing the **development and protection of digital infrastructure**, including cyber security and sustainability of the infrastructure itself, to the **processes and tools for data exploitation**, with particular reference to AI-based/enabled systems. Other relevant policy areas are **digital education and capacity building**, as well as the **digitalisation** of both the industry and public administrations (See the objectives identified and prioritised by the analysed countries in Figure 10). Based on this, four main clusters of objectives can be identified.

First, **infrastructure modernisation and connectivity** are highly prioritised, with reference to cloud and edge computing, 5G & 6G deployment and ensuring sustainability of the infrastructure. These priorities reflect wider objectives to strengthen Europe's digital foundation, recognising resilient and efficient digital systems as essential for competitiveness and security.

Second, **public administration and industry digitalisation combined with data security** also emerge as common objectives, requiring the development of digital service platforms and portals, as well as secure data management systems, while ensuring the protection of personal

data. Compliance with this dual objective of expanding digital systems while protecting privacy should be taken into account when developing new solutions and related governance.

Third, acknowledging the impact and potential of AI for the activities of both public and private stakeholders, policies prioritise the **integration of AI as a tool in various domains and sectors** (e.g. the Energy, Security, Climate resilience, Agriculture domains as mentioned in this study), as well as the **training of derived applications**, which requires a **significant amount of trusted sectoral data**. In this context, Earth Observation solutions can be instrumental in providing consistent and standardised datasets.

Finally, the **expansion of digital coverage and connectivity** also remains a high priority, with a focus on reaching underserved and remote areas, in combination with digital education and capacity-building, the latter also enabling the faster digitalisation of industrial and public entities' processes. An **opportunity for Earth Observation** arises in this context through its **integration into broader digital training agendas**, without remaining confined to experts.

Overall, the above-mentioned priorities align closely with the EU's "Digital Decade Policy Programme 2030", also due to the consideration of Digital Decade Strategic Roadmaps – among other policy documents – in the analysis, which sets out four pillars for the EU's digital transformation: enhancing digital skills, digital transformation of businesses through AI and cloud, secure and sustainable digital infrastructures, and the digitalisation of public services.[14]

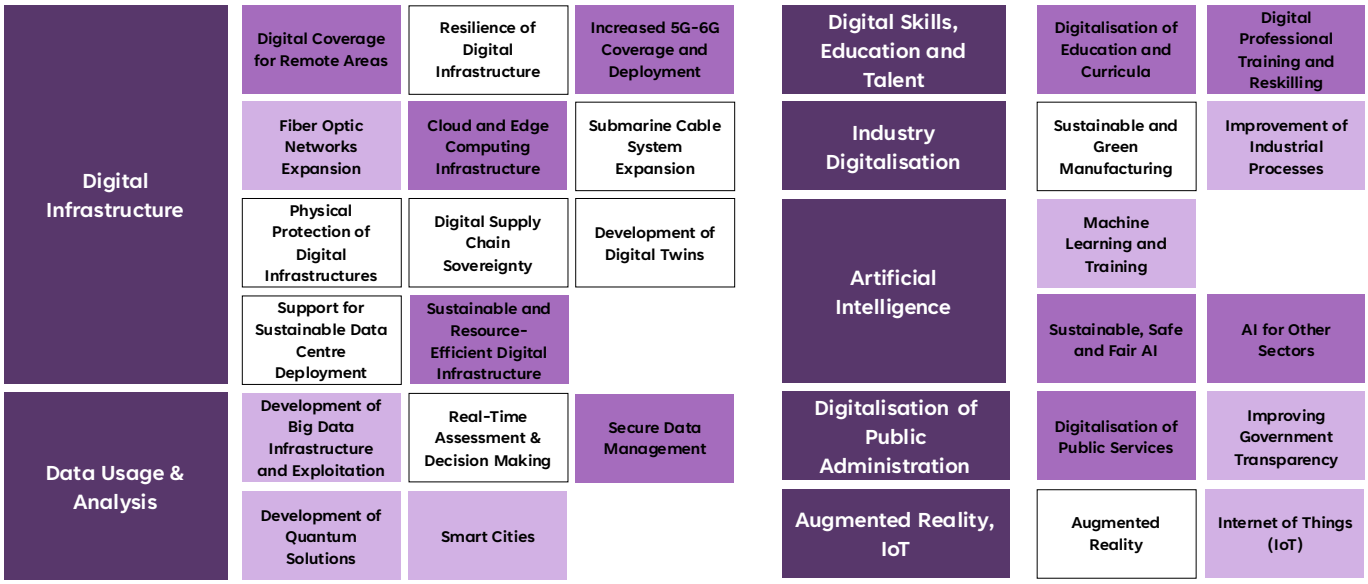
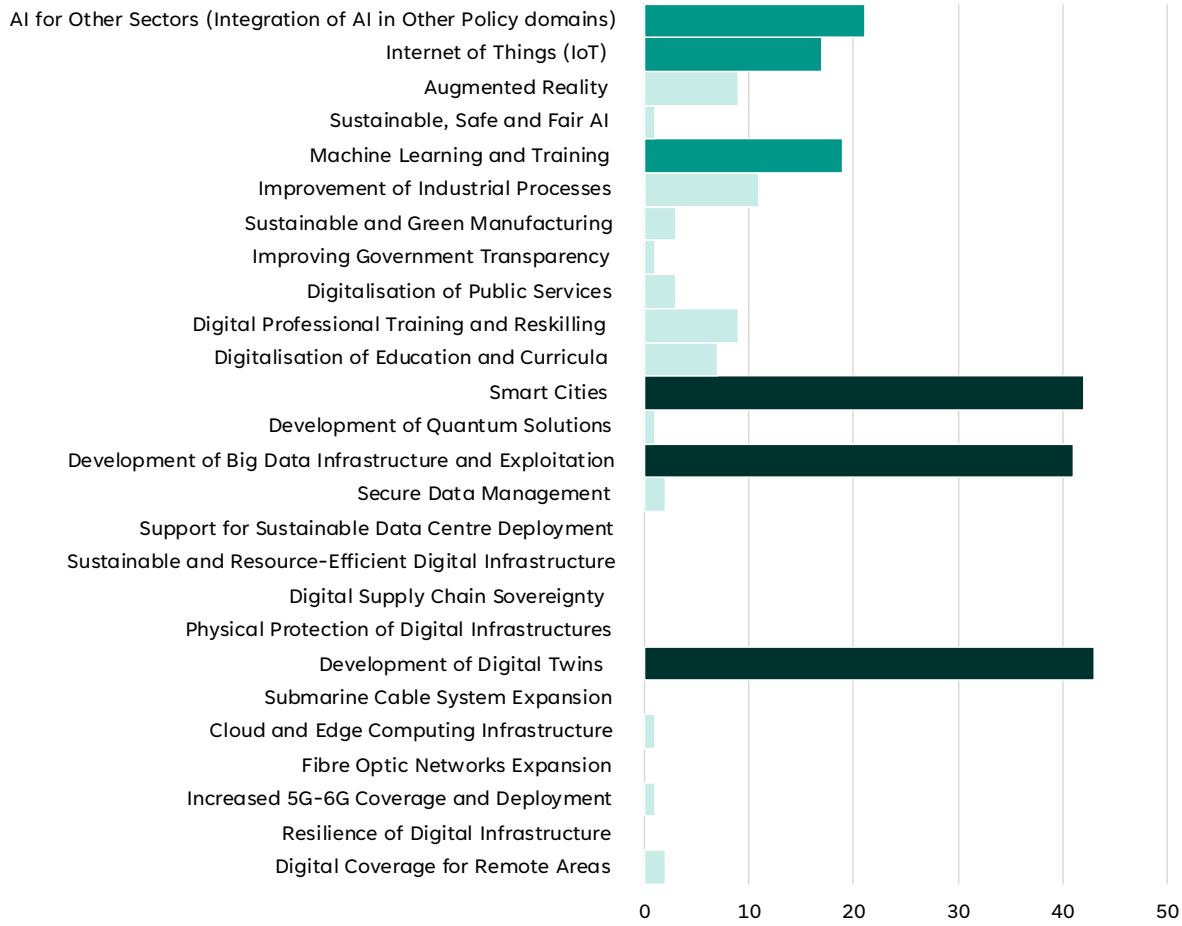


Figure 10: Digital Transition use cases prioritised by >65% (light purple) and >90% (purple) of analysed countries across the nine categories outlined on the left (dark purple)

[14] 'Europe's digital decade: 2030 targets' - European Commission

4.2 Earth Observation Projects Overview

The Earth Observation projects directly relevant for the Digital Transition domain (just under 250) are distributed across use cases as shown in Figure 11. The *Real-time Assessment and Decision Making* use case is added separately as 'outlier', being addressed by two thirds of the total identified projects, also considering the broadness of its scope and the consequent transversality to a plurality of other domains beyond the digital transition. The distribution below only includes projects explicitly mentioning one or more of these use cases within their objectives – it does not account for any indirect or potential relevance of projects for other use cases.



Outlier: *Real-time Assessment and Decision Making*, addressed by 2/3 of the Digital Transition projects (>160 units)

Figure 11: Distribution of Earth Observation projects across Digital Transition use cases, categorised using a three-tier scale indicating the number of projects per use case

Overall, projects are not distributed uniformly across Digital Transition use cases, with a strong concentration in the 'outlier' and 'broad-scope' use case of Real-time Assessment and Decision Making and other (significantly smaller) peaks aligning with a limited number of policy objectives. While the prominence of this use case reflects the strength of Earth Observation solutions in contributing to timely decisions through **near-real-time data and continuous monitoring**, the imbalance highlights the **need to broaden Earth Observation applications across more specific objectives of the digital transition agenda**, where feasible, which are not covered by screened projects. Nonetheless, Earth Observation solutions significantly contribute to the

development of applications in several use cases such as for smart cities, as well as to the development of big data infrastructures and related creation of digital twins. These are gathering increasing attention in policy documents and implementation strategies. Indeed, reviewed projects demonstrate the capacity of Earth Observation solutions for continuous monitoring and subsequent provision of data for urban management and modelling, with digital twins translating this into actionable insights for decision-making. In this context, significant overlap of projects across the smart cities and digital twins use cases confirms the **emergence of cities as pilot environments where Earth Observation data converges with other digital applications to support efficient and sustainable planning and resilient infrastructure**. According to the projects screened, urban contexts receive particular attention in areas such as smart applications and urban heat islands. However, the focus on smart cities also highlights the need to extend digital twin and smart applications beyond cities into regional and rural planning to ensure their benefits are realised beyond urban settings.

A **second set of use cases** (in orange in Figure 11) is served to a limited but still significant extent by Earth Observation projects, including the role of **Earth Observation as an enabler of digital tools based on AI and IoT applications**. While the use of Earth Observation in combination with these tools is much more significant, the projects were selected specifically to only highlight cases where Earth Observation serves as an enabler through the **provision of trusted, neutral and large volumes of data**.

The relevance of Digital Transition projects **across other domains** is overwhelmingly shaped by the projects aimed at **providing real-time assessment and contributing to decision making**. This overlap with other domains includes objectives in the areas of disaster management (e.g. early warning systems, natural hazard planning & response and particularly the case of rapid-onset events such as wildfires and floodings), infrastructure monitoring (incl. critical energy infrastructures), and crop health and yield forecasting. This shows the twofold nature of digital tools, being both the object of dedicated policy measures, as shown by their inclusion in Digital Transition policies, and a **tool to enable policy action in other domains**. The cross-domain relevance of Digital Transition projects with Agriculture & Food use cases, such as the use of AI-enabled digital tools for agriculture activities, should be understood in this context; reviewed projects confirm the increasing contribution of Earth Observation solutions to precision agriculture, supporting objectives such as irrigation optimisation and animal welfare.

Finally, significant cross-domain relevance of Earth Observation projects for **smart cities** is visible with the use cases on the **understanding and management of urban heat island effects**, in line with the integration of urban resilience challenges into smart city initiatives. In this context, while Earth Observation is already deeply embedded into climate hazard monitoring (e.g. flooding), its value can be maximised, as shown by reviewed projects, when its solutions also contribute **to long-term adaptation (and mitigation) through digital twins** (e.g. development of coastal zone simulations, installation of cooling infrastructure, formulation of greening strategies) **and to the energy transition** (e.g. supporting energy demand forecasting and building renovation planning). Moreover, some projects are already integrating Earth Observation within digital platforms and tools for citizen engagement to help public authorities co-develop climate mitigation and adaptation measures in line with European Green Deal objectives. This is the case, for instance, of observatories that integrate mobility, air quality, and healthy living data to guide the creation of carbon-neutral neighbourhoods, with a side impact also on monitoring compliance with climate-related goals.

4.3 Opportunity Analysis

The comparison of policy priorities with the ongoing and planned projects addressing them (See §§4.1 & 4.2) enables the identification of gaps in the contribution of Earth Observation solutions to Digital Transition policies, representing areas where current project implementation does not yet align with established priorities and related opportunities. The assessment of the technical feasibility of solutions filling existing gaps is beyond the scope of this study.

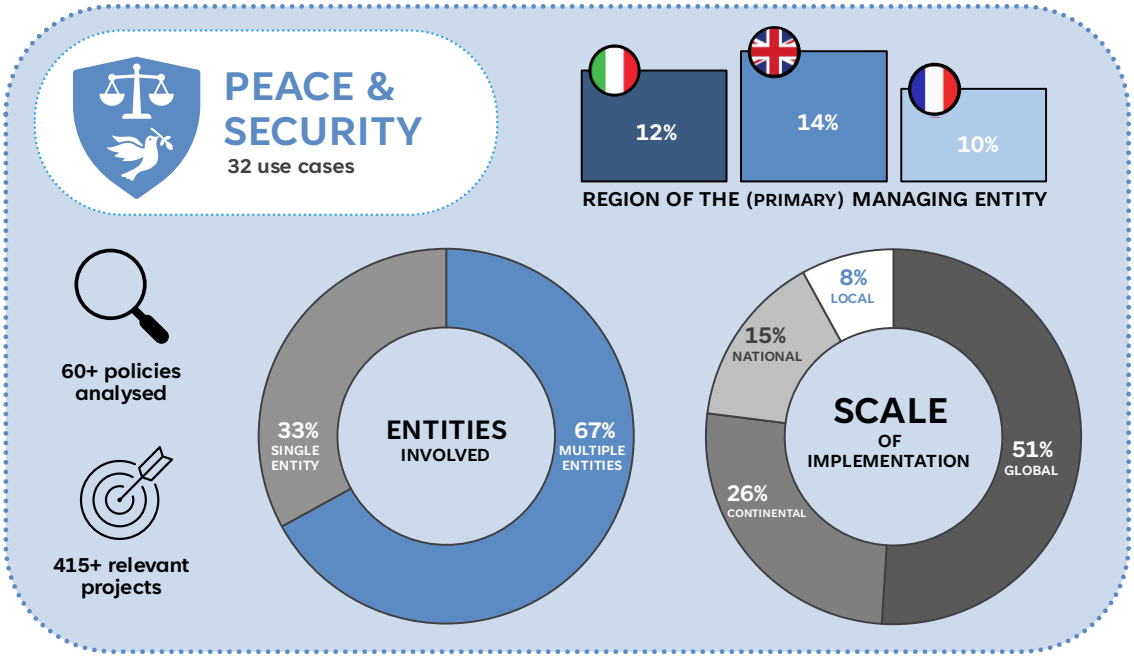


Overall, there is moderate alignment between policy priorities and the current distribution of projects in the Digital Transition domain, also **in view of the limited potential direct contribution of Earth Observation solutions to some use cases** (e.g. quantum computing or the development of cloud and edge infrastructure). This is due to a focus of the analysis on the sole contribution of Earth Observation solutions as enabling digital technologies (Earth Observation *for* Digital), excluding the cases where Earth Observation applications are the users of other digital tools, such as the above-mentioned cloud infrastructure. In this context, Earth Observation continues to function and is **gathering increasing interest primarily as a data source underpinning the development of digital applications**, such as those based on AI, with the potential of supporting the scaling of their socio-economic and environmental impact. This being the bigger picture, the contribution of Earth Observation solutions to several key digital transition objectives, such as the development of smart cities or digital twins to support policy action in other domains, is significant (See further in §4.2). Nevertheless, a few **main opportunities** should be recognised to increase the role of Earth Observation as a factor actively advancing the digital transition:

- *Enabling AI for other sectors* – In this top-priority policy area, representing one of the objectives but also a key driver of the Digital Transition, Earth Observation has potential to play a greater role in **enabling AI applications beyond the current level of data provision** for training, validating and feeding the models (e.g. Earth Observation feeding AI models for assessing plant physiology and disease, as derived by collected projects). This requires the integration with other data sources, thus developing and adhering to specific standards.
- *Digital Professional Training & Reskilling* and *Digitalisation of Education and Curricula* – As a key priority for developing the current and next-generation workforce, this objective is not yet fully addressed by Earth Observation projects, which are more focused on technical outputs. Instead, maximising the impact of such outputs would benefit from the integration of Earth Observation in training and education.
- *Increased 5G-6G Deployment* and *Digital Coverage for Remote Areas* – As shown by reviewed projects, Earth Observation supports the planning and optimisation of telecommunications networks by mapping terrain, infrastructure, and settlements to identify suitable locations for masts, antennas, or fibre deployment. It is also a valuable tool for detecting isolated or underserved communities, which supports targeted investments in rural and remote broadband and mobile coverage, while contributing to the promotion of network redundancy by helping to identify vulnerabilities and alternative routing options.
- *Improving Government Transparency* – High priority not only at the policy level, but also within the **political sphere**, currently served to a very limited extent by Earth Observation. The unlimited potential for the development of new projects in this area is also in line with the need to **tackle the disinformation threat** within the Peace & Security domain, where synergies can be found.
- *Sustainable and Resource-efficient Digital Infrastructure* – With growing data collection and processing needs, a focus on efficiency and sustainability to boost the development of the infrastructure can be enabled by Earth Observation.[15]

[15] See also on this topic the ESPI Report 'Data centres in space: orbital backbone of the second digital era?' (2025) – ESPI

5 PEACE & SECURITY



General insights

- Significant prioritisation of Civil Security, Law Enforcement and Diplomatic use cases, **beyond Defence**.
- High number of projects contributing to the protection of infrastructures and essential systems, early warning systems and disaster response.
- **Low integration of references to Earth Observation in policies**, especially compared to other domains.

5.1 Policy Landscape

The priorities within the Peace & Security domain, stemming from security and foreign affairs policies of European countries, can be categorised across four pillars: **law enforcement**, **international affairs**, **civil security & crisis management** – including related to natural disasters – and **defence** (See the objectives identified and prioritised by the analysed countries in Figure 12). Three main elements emerge from the analysis:

First, policies prioritise the enhancement of **security situational awareness** and **systems resilience**. While this involves defence-related objectives and instruments, significant attention is dedicated to natural and climate-induced stressors and crises. Still, peacebuilding remains the highest priority beyond the objectives related to tackling environmental issues. In this context, reliable information for managing risks to essential systems and infrastructures, as well as identifying vulnerabilities in energy and transport networks and strengthening public health protection is key. Solutions based on Earth Observation data, which already demonstrate their added value in closely related domains like climate resilience (See §6), can play a critical and increasing role in addressing these Peace & Security use cases.

Second, the strong emphasis placed on international and development cooperation, treaty verification and human rights, and peace promotion highlights the emergence of an autonomous policy domain, potentially under the '**Diplomacy**' title, increasingly requiring concrete tools for implementation. This is confirmed by consulted experts in the field (e.g. former and current ambassadors and technical personnel of international organisations), also highlighting the need for advanced solutions, including pilots based on Earth Observation data, producing evidence through trusted and neutral data in support of diplomatic action.

Finally, there is a high prioritisation of the emerging **disinformation threat**, where neutral observational technologies such as Earth Observation could play a key role in mitigating the societal, economic and environmental risks generated. While this is linked to the use of such technologies for fact-based diplomacy, its potential goes far beyond the diplomatic context.

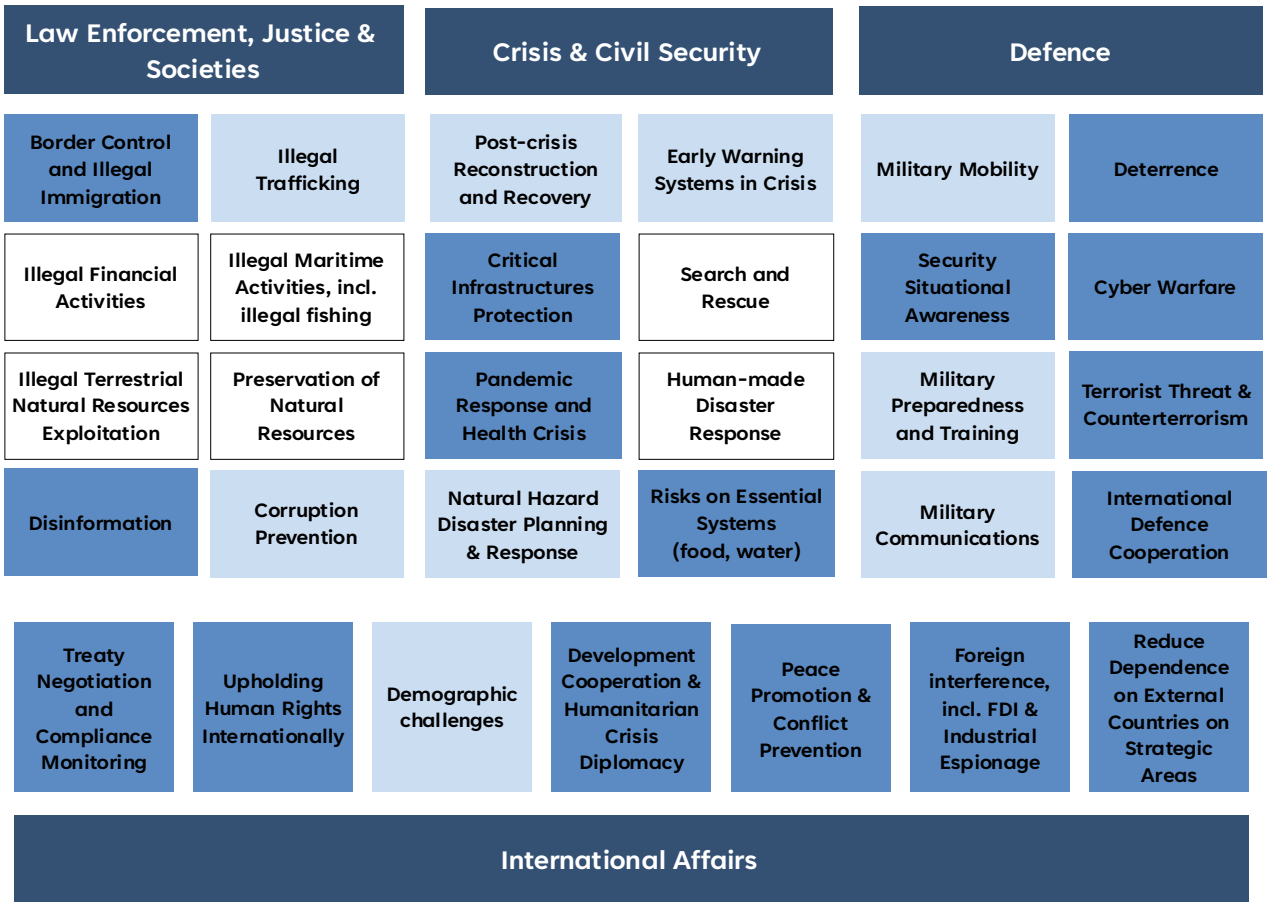
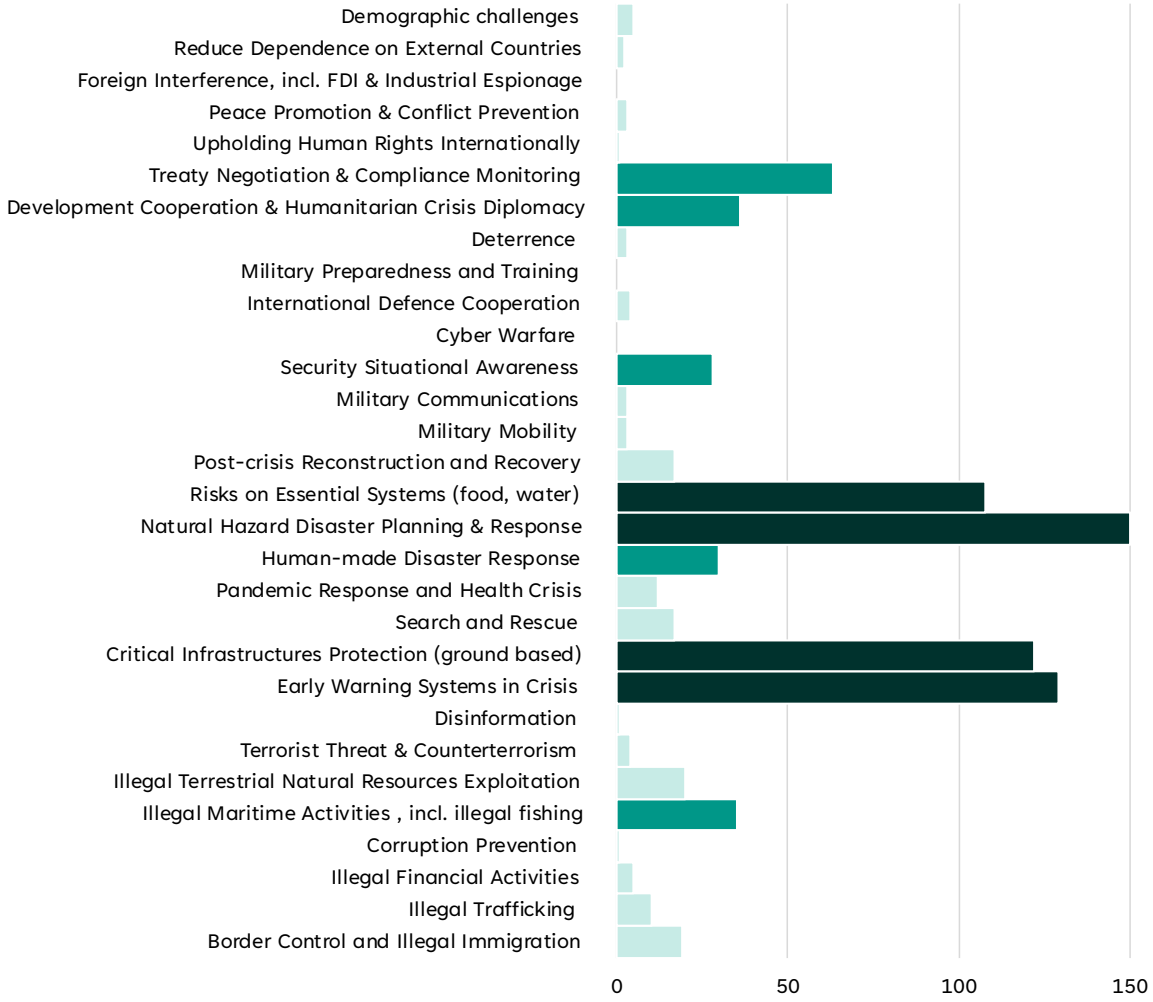


Figure 12: Peace & Security use cases prioritised by >65% (light blue) and >90% (blue) of analysed countries across the four categories outlined above (dark blue)

5.2 Earth Observation Projects Overview

The Earth Observation projects directly relevant to the Peace & Security domain (over 400) are distributed across use cases as shown in Figure 13. The *Preservation of Natural Resources* use case is shown separately as an 'outlier', as it accounts for more than half of the total relevant projects in the domain. This is due to its broad scope and strong overlap with other use cases, especially within the Climate Resilience domain. For this reason, the presence of this use case also **influences the high number of projects considered relevant for the Peace & Security domain**, which would be far more limited if this use case were to be excluded, as well as others related to climate resilience (e.g. *Natural Hazard Disaster Planning and Response*). The distribution below only includes projects explicitly mentioning one or more of these use cases within their objectives – it does not account for any indirect or potential relevance of projects for other use cases.



Outlier: *Preservation of Natural Resources*, addressed by more than half of Peace & Security projects (>250 units)

Figure 13: Distribution of EO projects across Peace & Security use cases, categorised using a three-tier scale indicating the number of projects per use case

Projects collected and analysed as relevant for the Peace & Security domain are constrained by the difficulty of achieving a full mapping due to limited public information on several security- and most defence-related Earth Observation solutions addressing the identified use cases (See further in §5.3). Nevertheless, the following considerations can be formulated, with particular focus on use cases where a more solid and reliable collection of projects was made.

Overall, projects are not uniformly distributed across Peace & Security use cases, even when defence-related ones are excluded. Indeed, the majority of projects include the preservation of natural resources among their objectives. While this objective reflects the well-established role of Earth Observation in the biophysical monitoring of ecosystems and natural resources, only further analysis and breaking this use case into sub-components (e.g. by type of resource addressed) would enable a more precise understanding of the role of Earth Observation in this area. **'Natural Resources'** emerged as an autonomous domain, including through consultation of experts in the field (i.e. international and European organisations dealing with security and environmental matters). This distinct domain also includes use cases related to natural resource exploitation and management, where Earth Observation solutions already contribute to detecting illegal logging, mining, fishing and pollution. The majority of Earth Observation projects supporting the detection of illegal exploitation also directly contribute to **treaty negotiation and compliance monitoring**, for instance by supporting the enforcement of the EU Deforestation Regulation (EUDR) – tracking forest change, monitoring protected areas, and verifying carbon offset commitments.

The challenge and opportunity in this area is the **uptake of Earth Observation beyond environmental governance to other treaty areas where trusted, verifiable data is equally essential**. Finally, in this context, the use of Earth Observation in **detecting illegal maritime activities is paving the way for other adjacent applications, such as illegal trafficking**.

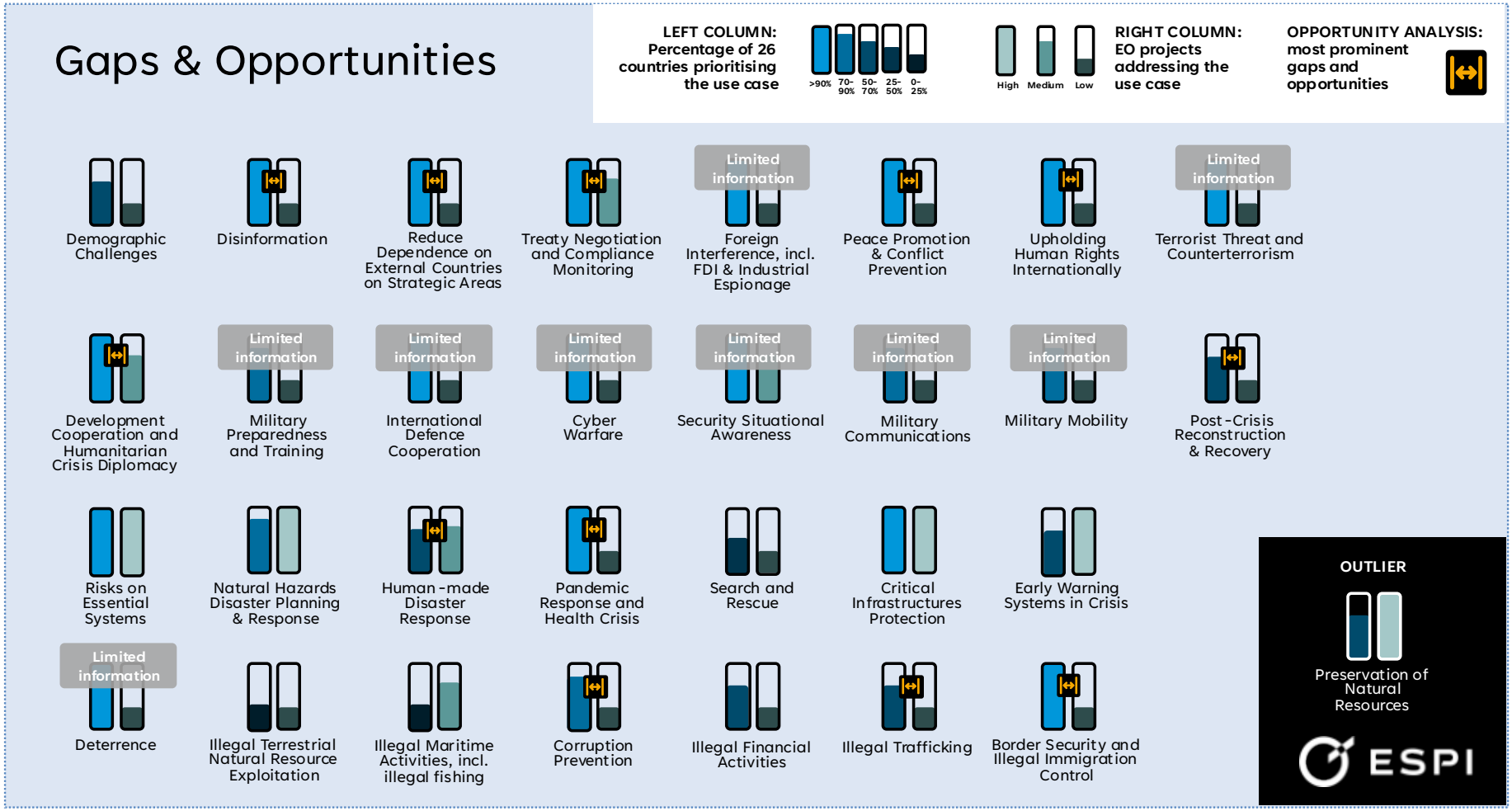
In parallel, **other use cases** related to disaster management, environmental monitoring, and infrastructure and systems protection – each with broad scope and **direct or indirect links to climate resilience** – are addressed by a significant number of projects. In this context, the focus on (natural) crisis response and early warning systems reflects the established role of Earth Observation in disaster management, with projects serving a dual-purpose: protecting ecosystems and the human-built systems that interact with or depend on them. Growing climate risks reinforce the importance of Earth Observation in disaster management and environmental protection. However, the strong **concentration of projects in these traditional applications should be combined with addressing other pressing priorities, such as conflict prevention, post-crisis reconstruction and recovery, treaty verification, and countering disinformation**.

A significant number of projects supporting the mitigation of risks for essential systems is also relevant to development cooperation initiatives and humanitarian crisis response, with **particular focus on strengthening food and water security across the African continent** and in other regions heavily affected by climate change, conflicts and socio-economic challenges. For instance, Earth Observation is used in support of food security by feeding AI models capable of distinguishing land-use patterns, assessing crop productivity, and enabling efficient precision agriculture, as demonstrated by projects appearing transversally across the Peace & Security, Digital Transition, and Agriculture & Food (incl. by ensuring food access to vulnerable populations) domains.

Further relevance of projects **across domains**, especially Peace & Security, Climate Resilience, and Agriculture & Food is visible in disaster-related use cases, as expected and with particular focus on rapid-onset events. Earth Observation projects provide decision-makers with critical tools to test scenarios, anticipate risks, and design more resilient systems before hazards occur. Apart from the monitoring and protection of infrastructures, limited overlap is observed with the Energy domain, highlighting the need for a more systematic approach to addressing priorities at the intersection of Security and Energy (e.g. supply diversification).

5.3 Opportunity Analysis

The comparison of policy priorities with the ongoing and planned projects addressing them (See §§5.1 & 5.2) enables the identification of gaps in the contribution of Earth Observation solutions to Peace & Security policies, representing areas where current project implementation does not yet align with established priorities and related opportunities. The assessment of the technical feasibility of solutions filling existing gaps is beyond the scope of this study.



When analysing the contribution of Earth Observation solutions to the Peace & Security domain, the apparent low alignment between policy prioritisation and Earth Observation project distribution should not be understood in general as an indicator of limited activity and use of Earth Observation. Indeed, several hard-security/defence-related use cases are served by Earth Observation applications, with related detailed **information not publicly available**, including straightforward acquisition of commercial data, and therefore beyond the scope of this study. For this reason, considerations on civil security, law enforcement, crisis management and international affairs will be primarily formulated for use cases where the contribution of Earth Observation can be reliably assessed.[16] This approach also has the potential to **shed light on the variety of security applications beyond defence, raising significant opportunities for expanding the impact of Earth Observation solutions**.

Overall, the concentration of projects in a limited number of use cases (See §5.2) reflects relatively strong alignment between policy prioritisation and Earth Observation activity in more traditional domains such as environmental protection and disaster management. Nevertheless, a notable gap is observed in other use cases, such as those related to Diplomacy, where policy ambition is high, but project activity remains limited. **Opportunities** arise in several areas:

- *Disinformation Threat* – Top priority at the policy level – and highly relevant for the political context and dynamics – the role of Earth Observation as an independent and neutral source of information is not yet leveraged at its full potential. This includes serving as a tool for fact-based diplomacy, thus preventing escalation and actively contributing to broader objectives such as conflict prevention and peace promotion. Imagery intelligence for anomaly detection and hotspot mapping of potential training camps or abnormal activities is also growing in the context of counterterrorism activities.
- *Development & Humanitarian Crisis Cooperation*, and *Upholding Human Rights* – Limited number of projects addressing these policy priorities, as also confirmed by experts and actors in the field (e.g. current and former ambassadors). The independence and neutrality of Earth Observation data can be further leveraged to verify *in situ* data and/or provide an established source of evidence to parties. Projects are already contributing to the identification of mass graves, detection of settlement changes, and fire detection in settlements.
- *Border and Immigration Control* – While the lack of publicly available information affects the assessment of the contribution of Earth Observation projects to this use case (See the 'caveat' above), which may therefore suggest low alignment with policy prioritisation, several projects can be identified that support authorities in detecting illegal transport of goods and people to prevent illicit trafficking and irregular migration, as well as contributing to search and rescue operations. Nevertheless, to maximise the impact of such solutions there is scope to extend access, where relevant, beyond government authorities to NGOs, given their crucial role in crisis situations.
- *Pandemic and Health Crisis* – Recent prioritisation in policy has been accompanied by growing activity at the project level. In particular, projects leverage Earth Observation data for monitoring and alert systems for vector-borne or waterborne diseases, as well as using hyperspectral data to understand how environmental factors influence the spread of viruses.[17]

[16] See the definition of these categories in the ESPI Report 'Space in Support of Security Missions' (2022) - ESPI

[17] In this regard, See the ESPI Report 'COVID-19 and the European Space Sector' (2020) - ESPI, where broader consideration on the role of Earth Observation, and more broadly space, for pandemic crises response and healthcare are formulated.

UPTAKE IN PRACTICE – EUROPEAN MARITIME SAFETY AGENCY

About the European Maritime Safety Agency

The European Maritime Safety Agency is one of the European Union's decentralised agencies. Based in Lisbon, the Agency's mission is to serve EU maritime interests for a safe, secure, green and competitive maritime sector and act as a reliable and respected reference in the maritime sector in Europe and worldwide.

The responsibilities of EMSA's Earth Observation services include operating the CleanSeaNet service, the European satellite-based pollution monitoring and vessel detection service, serving as the Entrusted Entity for the implementation of the Copernicus Maritime Surveillance service on behalf of the European Commission, supporting emergencies at sea, and supporting Frontex in border control activities.

Data sources – To provide its Earth Observation products in quasi-real time, EMSA relies on Synthetic Aperture Radar (SAR) and optical data. Commercial platforms and Copernicus Sentinel-1 data are downlinked to ground stations operated by EMSA's contracted service providers, processed and delivered to EMSA's systems. Relevant information is provided to end-users through Alert reports, a web portal, mobile applications, or secure system-to-system channels.

Benefits of Copernicus data and solutions

User Maturity Level 5 – Operational User[18]

Several examples of how Earth Observation solutions specifically based on Copernicus data are supporting the operational activities of EMSA are provided below, complemented by needs for improvement.[19]

- **Copernicus** is a **key asset** for **pollution detection** at sea (e.g. oil spills),[20] with
- **Full operational** pollution monitoring service and **routinely integrated into end-user workflows**, supporting quasi-real-time situational awareness and decision making.
- **Time-sensitive activities enabled** through the delivery of satellite images analysed and integrated with EMSA datasets **within 20 mins from acquisition**.
- **Strong contribution** to **EU policy implementation and monitoring** (e.g. Integrated Maritime Policy, Environment Policy, Common Security and Defence Policy, Arctic Policy, etc.).

Outstanding needs for future improvements

- Further innovative integration of Earth Observation with other data sources (e.g. vessel traffic information, user intelligence such as behavioural best practices)
- Availability of very high-resolution imagery for search-and-rescue operations

[18] Full integration of Copernicus-based solutions into their standard processes, where the necessary resources, including staff, budget, and facilities, are already allocated or readily deployable, as assessed by EMSA.
[19] The assessment is based on consultations with EMSA and subsequent validation by EMSA of the information hereby provided.
[20] EMSA assessed the effectiveness scoring it 5 on a scale of 1 (minimum) to 5 (maximum). A broader explanation of the use of CleanSeaNet for oil spill detection and monitoring is provided for instance in Clean Seas in the Mediterranean, EARSC, 2022.



UPTAKE IN PRACTICE – SPANISH CIVIL GUARD

About the Spanish Civil Guard

The Spanish Civil Guard is a law enforcement agency overseen by the Spanish Ministry of the Interior and the Ministry of Defence, with approximately 80,000 employees.

The **responsibilities** of the Civil Guard include the provision of maritime security services, such as border control, coastal and environmental protection, anti-smuggling operations, and maritime search and rescue missions, ensuring the safety of national waters and compliance with maritime laws and regulations.

Data sources – The Civil Guard uses Earth Observation data including through the European Maritime Safety Agency (EMSA) and Frontex, which provide information based on Earth Observation data in response to specific requests formulated by the Civil Guard. EMSA acts as the entrusted entity for the Copernicus Maritime Surveillance (CMS) service. Once data has been retrieved and processed by EMSA, it is delivered to the Civil Guard through secure web portals, mobile applications, or encrypted system-to-system connections.

Benefits of Copernicus data and solutions

User Maturity Level 4 – Early Adopter[21]

Several examples of how Earth Observation solutions specifically based on Copernicus data are supporting the operations of the Civil Guard are provided below, complemented by needs for improvement:[22]

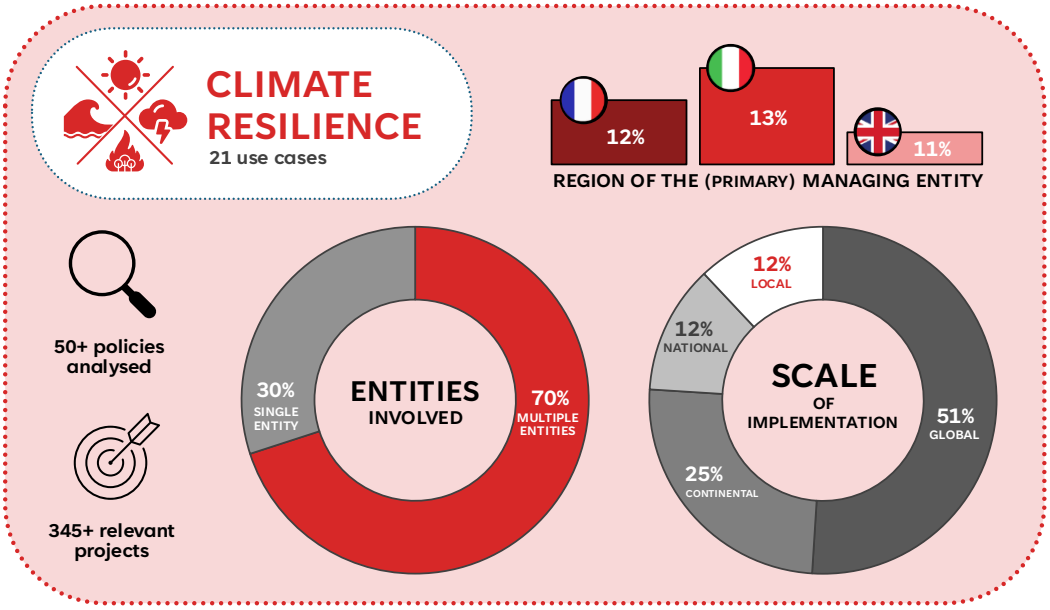
- **More than 1 day saved and significant effectiveness gained per average investigation** compared to traditional methods across three operational units of the Civil Guard: Irregular Immigration Investigation, Environmental Protection and Risk Analysis.[23]
- **Significant cost reduction** due to more limited deployment of equipment and long-distance travel, primarily benchmarked against the cost of offshore patrol vessel operations.
- **Early detection of oil and chemical spills**, including the source. Monitoring of the direction of the spill, assessment of response operations and gathering of legal evidence. Successful examples include satellite imagery used to identify suspect vessels in the case of a chemical spill off the Mediterranean coast of Spain in 2024, also resulting in the closure of affected beaches.
- **Enhanced compliance with EU and national laws** against Illegal, Unreported, and Unregulated (IUU) fishing through constant & large-scale monitoring, and vessel detection. Overall, increased surveillance capacity, concrete evidence and enhanced transparency also in support of international cooperation against illegal fishing and for environmental protection.

Outstanding needs for future improvements

- Cross-sector coordination of Earth Observation product provision across law enforcement, civil protection, and defence to improve efficient use thereof
- Enhanced frequency of revisit, resolution of images and waiting time from request[24]

[21] High confidence in Copernicus-based solutions; actively integrating them into internal processes, as assessed by the Civil Guard.
[22] The assessment is based on consultations with the Civil Guard and subsequent validation of the information hereby provided.
[23] The three Civil Guard units consulted assessed effectiveness scoring it 4 or 5 on a scale of 1 (minimum) to 5 (maximum).
[24] The latter refers to delivery of Earth Observation insights by agencies such as EMSA and Frontex to the request of the Civil Guard.

6 CLIMATE RESILIENCE & EMERGENCY MANAGEMENT



General insights

- **Top policy priorities** are related to the **Land** ecosystem.
- **Most Earth Observation projects** are related to the **Maritime & Freshwater,** and **Land** ecosystems.
- **Highest percentage of projects conducted by consortia** and **highest percentage of civil institutional** activities (28%), alongside the Agriculture & Food domain.
- **A significant share of projects** are coordinated by nations in **Western (37%) and Southern Europe (30%),** where **flooding and wildfires** emerged as key priorities.

6.1 Policy Landscape

Policy priorities in the Climate Resilience and Emergency Management domain span across all **ecosystems**, i.e. land, maritime, freshwater, atmosphere, complemented by **biodiversity** and several use cases aggregated under the category of '**first level (climate) impacts**', such as extreme temperature and sea level rise (See the objectives identified and prioritised by the analysed countries in Figure 14). The policy analysis might underestimate the prioritisation of natural disasters which are not induced by climate factors (e.g. earthquakes), due to a primary use of climate resilience policies in the research.

In particular, **biodiversity is emerging as a key priority** within climate policies of European countries, together with associated use cases (species monitoring, reduction of habitat stress and lowering of the occurrence and impact of pests and diseases). This should be understood as a trend specific to the European region, which is not similarly reflected in the priorities of the broader Mediterranean region, including North Africa and Middle Eastern countries.[25] In parallel,

[25] This finding was developed through additional work conducted by ESPI in 2024 on the topic of Space for Climate Resilience and Food Security in the Mediterranean region.

land-related objectives are also strongly prioritised, in line with the priorities of other Mediterranean countries.

Use cases related to the **freshwater** ecosystems are also highly prioritised and represent the key emerging priority of the climate resilience domain more broadly, especially water quality, scarcity, and degradation. This aligns with the significant policy focus on **flooding**, whose first and most severe impacts include the reduced availability of drinking water and disruptions to irrigation systems. Alignment with the prioritisation of the water body quality use case in the Agriculture & Food policy domain can also be observed.

Finally, **heatwaves** are highlighted as a common priority among the analysed countries, with particularly **high concern for urban areas**. This aligns with the strong cross-domain relevance of the urban heat island challenge across Mediterranean countries, extending beyond Southern Europe to include North African and Middle Eastern states.

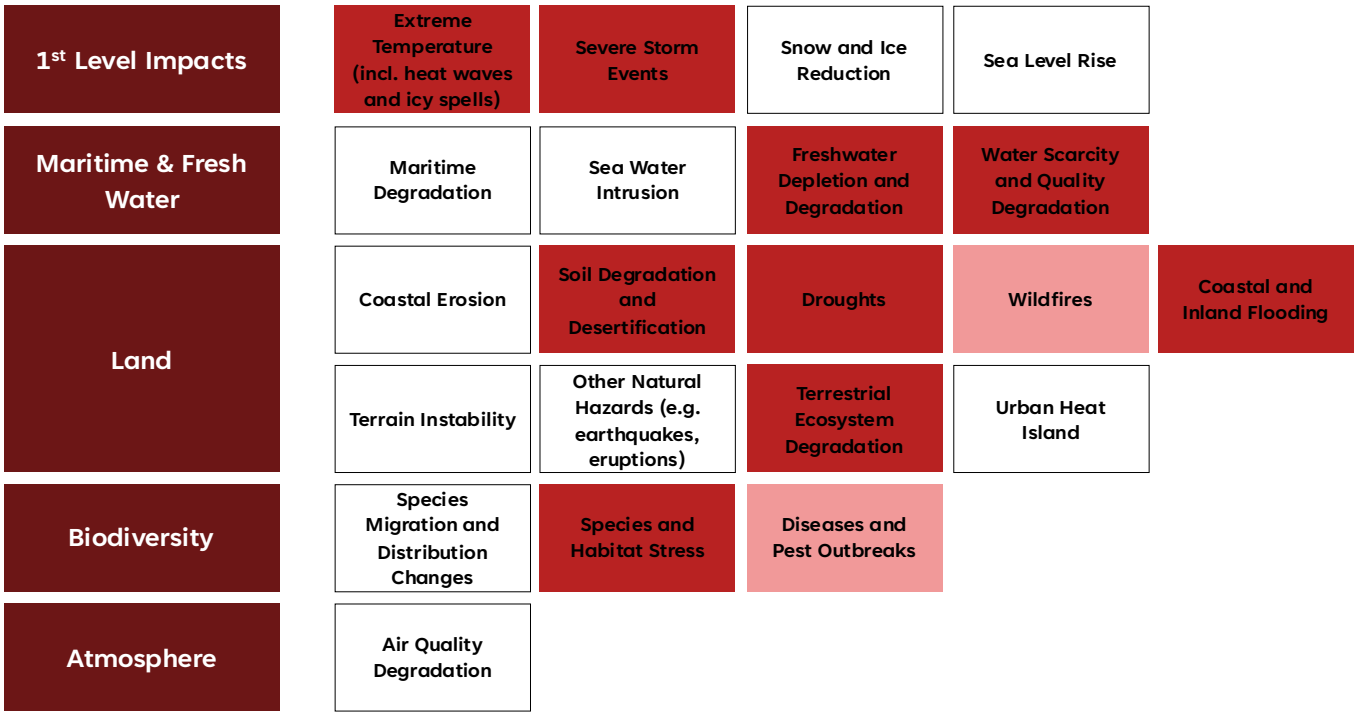


Figure 14: Climate Resilience & Emergency Management use cases prioritised by >65% (light red) and >90% (red) of analysed countries across the five categories outlined on the left (dark red)

6.2 Earth Observation Projects Overview

The Earth Observation projects directly relevant for the Climate Resilience & Emergency Management domain (over 350) are distributed across use cases as shown in Figure 15. The distribution below only includes projects explicitly mentioning one or more of these use cases within their objectives – it does not account for any indirect or potential relevance of projects for other use cases.

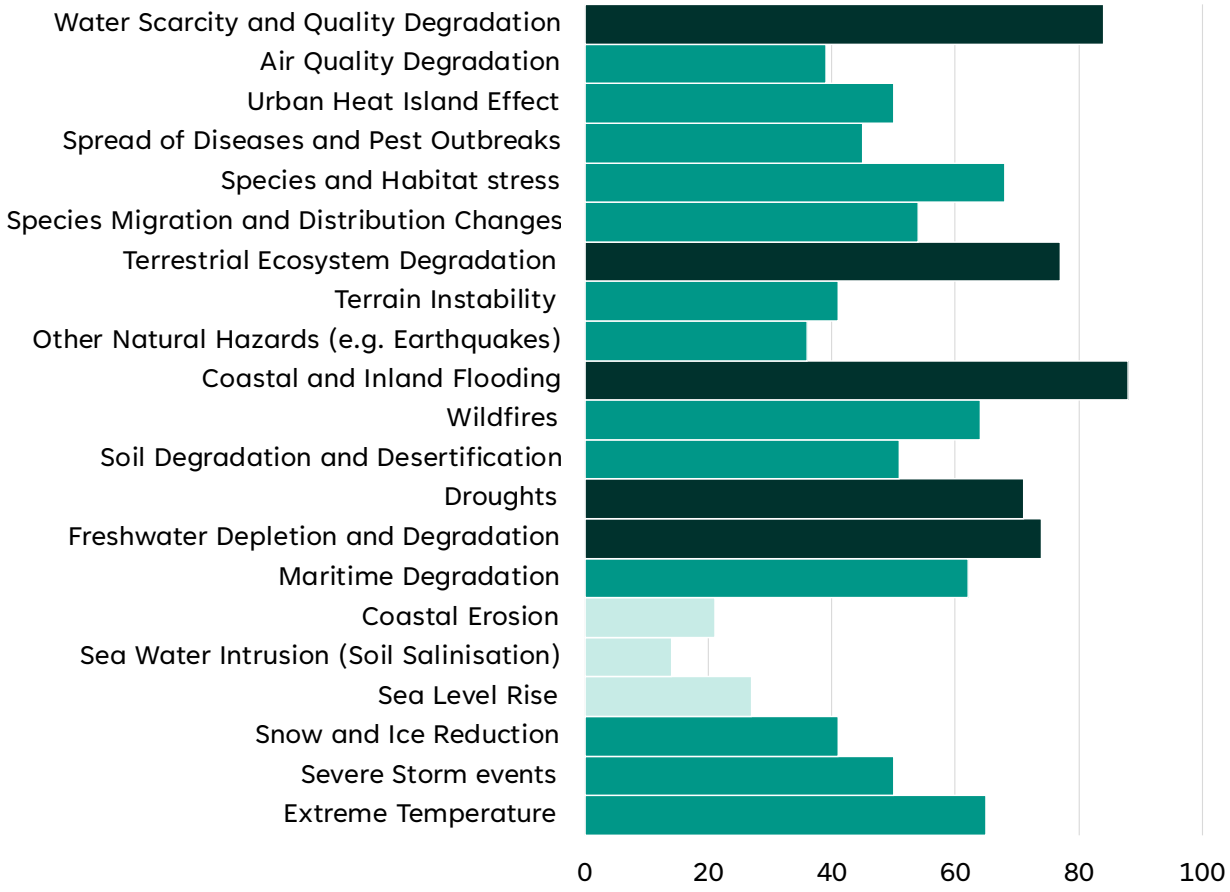


Figure 15: Distribution of Earth Observation projects across Climate Resilience & Emergency Mng use cases, categorised using a three-tier scale indicating the number of projects per use case

Overall, projects are distributed quite uniformly across use cases, indicating a **systematic and comprehensive contribution of Earth Observation solutions to Climate Resilience & Emergency Management** priorities, with lower numbers of projects in more niche use cases (e.g. coastal erosion, sea level rise), which are nevertheless indirectly addressed – even if not as the primary goal of the project – by other available Earth Observation solutions. This can represent a solid basis to further support domain objectives, especially by **expanding the most successful small-scale projects** currently ongoing, including through dedicated climate resilience and emergency management resources.

More specifically, flooding and storms are both significantly addressed by Earth Observation projects, with storms representing one of the main causes of both inland and coastal floods. This shows a **coordinated approach to addressing both the causes and effects of climate impacts**. Nevertheless, the number of Earth Observation projects supporting flood monitoring is significantly higher than those addressing storm events, indicating an opportunity to further

invest in prevention, forecasting and immediate response by leveraging solutions already available for post-disaster activities, thus increasing the opportunities to reduce damages. More broadly, this highlights a need to move increasingly **from a predominant focus on monitoring functions – typically associated with climate mitigation – towards applications that enable proactive planning and adaptation**. In this context, the products and solutions delivered by the European Organisation for the Exploitation of Meteorological Satellites (EUMETSAT) are key assets in support of climate resilience policies, as also recognised in their text through the frequent explicit references to EUMETSAT (See §8.1). Indicators of the recommended transition from post-disaster monitoring to prevention and response can be derived from the (although limited) number of projects relevant to both the Climate Resilience use case of *Coastal and Inland Flooding* and early warning objectives (Peace & Security) or real-time decision making (Digital Transition).

Moreover, flood risks are further compounded by drought, which reduces soil absorption capacity and increases runoff during heavy rainfall.[26] The IPCC classifies this as a compound event, where a climate-driven precondition exacerbates subsequent impacts. Projects addressing both drought and flooding illustrate how Earth Observation is **capturing these interlinked dynamics and supporting structured risk management**. Similar considerations apply to the relationship between wildfires and extreme temperatures, and between water scarcity, freshwater depletion and soil salinisation, the latter requiring increasing attention due to the rising policy concerns, especially in the Mediterranean region.[27]

When considering the relevance of Climate Resilience & Emergency Management projects **across other domains,** Agriculture & Food and Peace & Security emerge as the most directly connected. In the case of agrifood, climate resilience projects are relevant across many use cases, especially those related to soil and water, in particular when addressing floods, drought and freshwater degradation. In contrast, the relevance of climate resilience projects in the security domain is less distributed, in fact mainly relating to natural resource protection and disaster management, but also development cooperation. This highlights **opportunities to strengthen diplomatic action by leveraging the established Earth Observation solutions for climate and disaster response**.

Although more limited, a contribution of climate resilience projects to objectives in the Energy and Digital Transition domains is also observed, particularly regarding infrastructure monitoring and protection, and renewables deployment. For instance, Earth Observation solutions support renewable energy planning by monitoring snowmelt for hydropower or map thermal stress to guide the deployment of solar energy infrastructure.

[26] IPCC Working Group 1: The Physical Science Basis, Chapter 8: Water Cycle Changes, 'FAQ' – IPCC
[27] As confirmed by additional work conducted by ESPI in 2024 on the topic of Space for Climate Resilience and Food Security in the Mediterranean region.

6.3 Opportunity Analysis

The comparison of policy priorities with the ongoing and planned projects addressing them (See §§6.1 & 6.2) enables the identification of gaps in the contribution of Earth Observation solutions to Climate Resilience & Emergency Management policies, representing the areas where current project implementation does not yet align with established priorities and related opportunities. The assessment of the technical feasibility of solutions filling existing gaps is beyond the scope of this study.



Overall, strong alignment is observed between policy objectives and ongoing and planned projects, which particularly stands out when compared with other analysed domains and is also reflected in a broad and structured integration of references to Earth Observation in climate policies (See further in §§8 & 8.1). This results in very few gaps, while still leading to **significant opportunities**.

First, there is an opportunity to leverage existing Earth Observation programmes to deliver positive impact in neighbouring regions facing similar climate resilience challenges and other crises,[28] including as a tool for development cooperation or broader diplomatic action. This would also increase the contribution of Earth Observation projects to Peace & Security priorities, particularly those linked to diplomacy (e.g. peace promotion and conflict prevention, upholding of human rights, treaty negotiation and compliance monitoring, detecting of illegal natural resources exploitation). Second, there is an opportunity to scale existing projects, with Earth Observation becoming a primary tool not only to monitor climate impacts, but also to prevent and act upon them. This includes **leveraging climate-related resources to expand successful pilots into operational solutions and enabling technology**.

More **specific opportunities** can be identified in the following use cases (or clusters thereof):

- *Water-related use cases* are highly prioritised and already well represented in projects, underscoring the need to maintain this focus in future initiatives. At the same time, there is a clear opportunity in this area to expand relations and partnerships with Mediterranean countries. Indeed, water and freshwater security are also top priorities in this region, but remain insufficiently addressed by current space-based projects.[29]
- *Wildfires* – Identified as a key priority, particularly by Southern European countries, also with a growing relevance for international cooperation efforts. While already addressed by a significant number of projects, data and expert interviews highlight an opportunity for further uptake of Earth Observation solutions in this area, especially compared to other top priority use cases such as flood prevention and response. Given the rising frequency and severity of wildfires, stronger attention is drawn to this use case by policies, resulting in increased opportunity for Earth Observation to deliver and show its potential.
- *Diseases and Pest Outbreaks* and *Species and Habitat Stress* – Despite a strong policy prioritisation as part of the Biodiversity cluster, only a limited number of projects address this use case compared to other priorities. Transversal policy relevance of the biodiversity topic to the Agriculture & Food domain could be leveraged to attract resources towards initiatives contributing to both conservation and adaptation strategies.

[28] As derived from additional work conducted by ESPI in 2024 on the topic of Space for Climate Resilience and Food Security in the Mediterranean region.
[29] *Ibid.*



UPTAKE IN PRACTICE – CZECH FORESTRY INSTITUTE

About the Czech Forestry Institute

The Czech Forestry Institute (NLI) is a government entity operating **under the Ministry of Agriculture** of Czechia, with approximately 430 employees.

The **responsibilities** of NLI include the implementation of the National Forest Inventory, the monitoring of forest health and species composition, and the development and release of forest typology maps to support sustainable forest management and policy planning.

Data sources – To perform its activities, the NLI relies on multiple **sources of imagery data**, including Copernicus (especially Sentinel-2, with potential for CHIME and ROSE-L) and Landsat, alongside commercial data provision (e.g. Planet) complemented by airborne surveys for higher-frequency data. Cloud computing and multi-source machine learning tools are also employed to identify forest health trends and develop species composition maps.

Benefits of Copernicus data and solutions

User Maturity Level 4 – Early Adopter[30]

Several examples of how Earth Observation solutions specifically based on Copernicus data are supporting the operational activities of NLI are provided below, complemented by needs for improvement:[31]

- **10-20% increase in area coverage** and **3 to 5 days saved** when conducting forest monitoring activities (e.g. harvesting detection).
- **10-20% increase in pests & diseases hotspot detection** for forest value preservation.
- **More than 1 week saved in species classification** compared to traditional methods.
- **Significant efficiency improvement on regulatory reporting procedure,** e.g. pursuant to the European Union Deforestation Regulation (EUDR).[32]

Outstanding needs for future improvements

- Higher data frequency and integration of datasets (both Earth Observation and beyond) to overcome limitations like cloud cover and to capture the full forest structure
- Enhanced seasonal analysis and classification methods to account for natural vegetation changes
- Higher spatial resolution for the identification of small-scale activities (e.g. illegal logging, for which Earth Observation is currently not used) and routine management of forest units of approx. 0.5ha
- Products (e.g. forest maps) compatible with standards outlined in national & EU legislation, incl. verified information accuracy and due diligence procedures ensuring legal admissibility
- Capacity to directly measure forest structure (e.g. canopy height) and biomass (e.g. stock volume, aboveground biomass) from space for the assessment of forest parameters[33]

[30] High confidence in Copernicus-based solutions and actively integrating them into internal processes, as assessed by the NLI.
[31] The assessment is based on consultations with the NLI and subsequent validation by the NLI of the information hereby provided.
[32] The NLI assumes this improvement (related to time and other resources) scoring it 4 on a scale of 1 (minimum) to 5 (maximum).
[33] In this context, the NLI explicitly mentions the potential contribution of the recently launched ESA Biomass mission.

7 INSURANCE

Unlike the domains directly addressed by specific public policies, aspects concerning finance and insurance activities are mostly referenced within each sectoral policy (e.g. energy policies), rather than in standalone 'Finance & Insurance' policies at the national level. In this context, instead of extracting and aggregating finance- and insurance-related elements from each sectoral policy – an approach that would risk combining inconsistent data from policy documents across different domains into a potentially misleading single picture – a different methodological approach was adopted. (See further in Annex A – Extended Methodology).

The **insurance market was chosen as capable of representing a consistent share of private-sector interest in addressing several priorities found in the other selected policy domains** (e.g. energy, climate resilience, digital transition). Indeed, the analysis of the insurance market can help to identify use cases where private and public interests align (e.g. protecting critical infrastructure from floods) and where resources could therefore be combined to further develop Earth Observation solutions addressing issues at stake. The outcomes are presented in the following sections.

7.1 Insurance segments analysis

The analysis of the different insurance market segments (See Figure 16) considered products covering both individual assets, i.e., Property and Motor (in *orange* and *yellow*), and commercial activities, especially Agriculture, Transport & Logistics, Construction and Energy (in *blue*). These categories were drawn directly from the review of 104 selected insurance companies, with a higher level of detail for Property and Motor products, while only the three overarching categories of Natural Hazards, Human-made and Operational Risks (system failures along ordinary operations) were assessed for the selected businesses.

Overall, this brief overview of the non-life insurance market in Europe shows a variety of opportunities to expand the use of Earth Observation services to support and complement insurance products. Indeed, there is significant **alignment between the risks insured and both established and emerging Earth Observation applications** in the fields of climate resilience and emergency management, security, precision agriculture, and continuous monitoring of transport routes, construction sites and energy infrastructure.

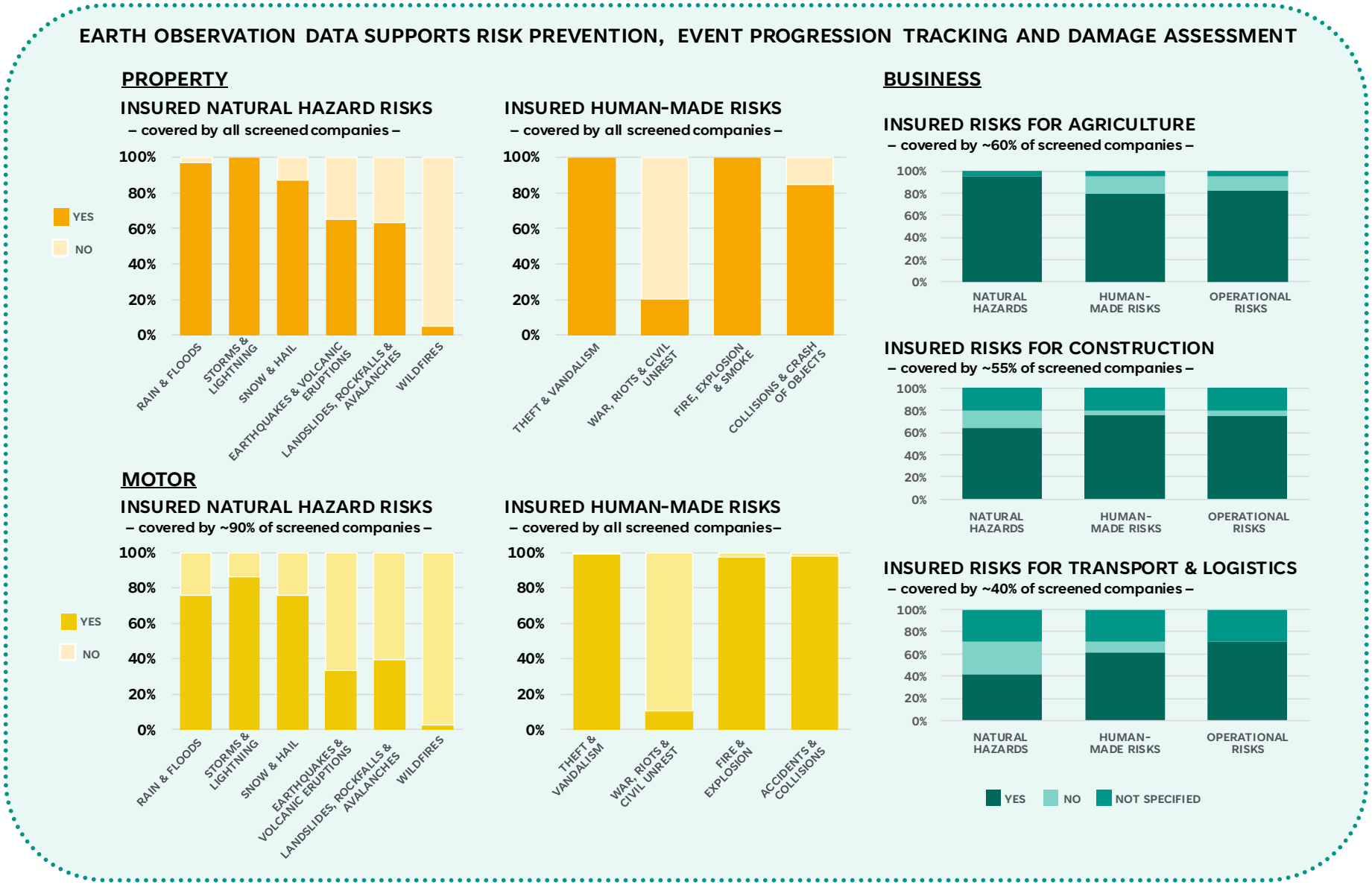


Figure 16: Insurance Segments Analysis

7.2 Public & Private interests combined

Additional useful insights for orienting the development, tailoring and specific application of Earth Observation solutions can be derived by matching the primary object of insurance products with main policy priorities at the national level, as outlined in strategic documents and presented in the previous domain analysis. In the context of Climate Resilience (See further in §6.1), there is a strong convergence of private and public interest in the case of severe weather events and their primary impacts, particularly coastal and inland floodings. Similar considerations apply in the case of wildfires.

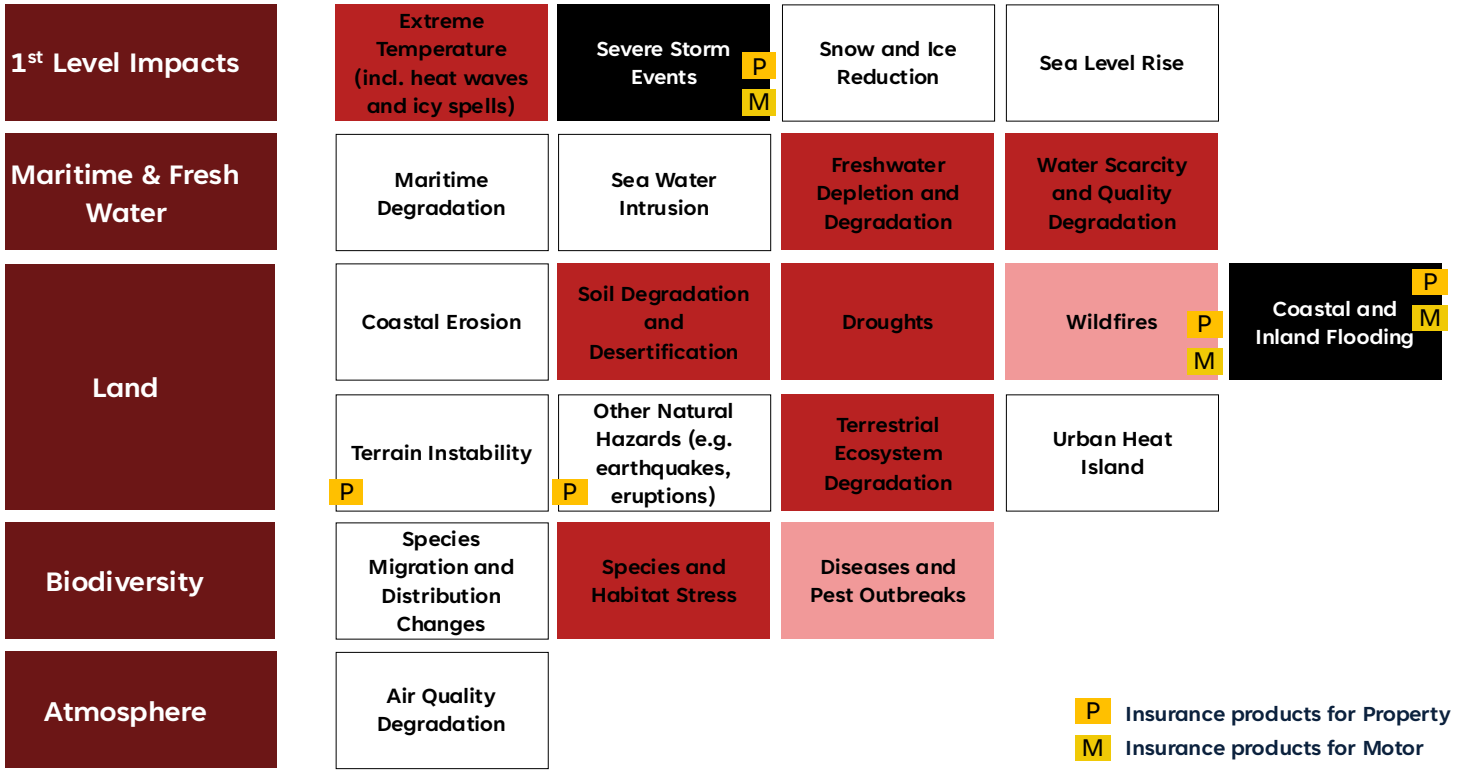


Figure 17: Combination of national climate policy priorities and main insurance solutions; Climate Resilience & Emergency Management use cases prioritised by >90% (red) of analysed countries across the five categories outlined on the left (dark red) and use cases where policy priority aligns with insurance market segments (black).

Several use cases in the Peace & Security domain (Figure 18, See also in §5.1) also show a high-level of convergence of public and private interest. In particular, this is seen in the area of civil security, particularly in the protection of critical infrastructure (assets, networks) and essential systems (services, functions), as well as in the (natural) disaster response domain, thus confirming the more detailed breakdown outlined in Figure 18.

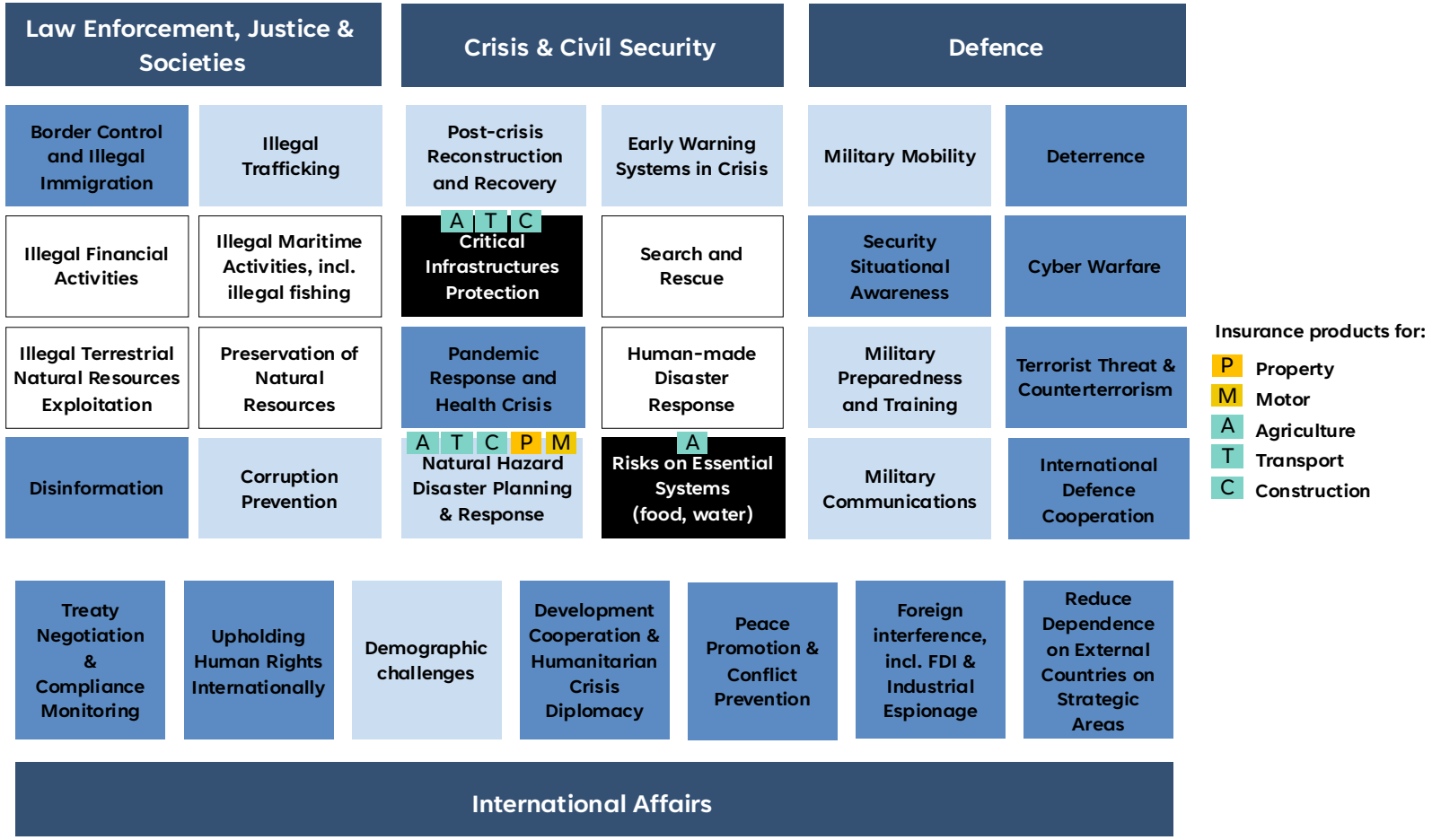


Figure 18: Combination of national peace & security policy priorities and main insurance solutions; Peace & Security use cases prioritised by >90% (blue) of analysed countries across the four categories outlined above (dark blue) and use cases where policy priority aligns with insurance market segments (black).

Transversally, value is seen in tailored services that support both policy implementation – across the whole policy cycle – and insurance, moving beyond a primarily monitoring function and becoming fully integrated in operations. More broadly, while the focus of high-level political debate is shifting towards defence applications of space assets, longer-term policymaking and the insurance market maintain a clear focus on resilience and civil security applications, justifying the efforts to ensure continuity and reinforcement of current Earth Observation applications.

7.3 Opportunity Analysis

Projects designed specifically for insurance purposes are still rare, although this finding might be biased by a preference for direct image acquisition for insurance purposes over the development of dedicated services. Nevertheless, projects not directly aimed at supporting risk insurance but addressing use cases overlapping with covered risk categories may hold relevance for the insurance sector. Taking this into account, opportunities were identified in domains where policy and insurance interests tend to align. The graph below aggregates results for a selection of use cases across the Climate Resilience and Peace & Security domains also relevant for Insurance The distribution of Earth Observation projects across these eight use cases is shown in Figure 19.

Overall, a partial alignment is observed between national policy objectives and main insurance market segments, with four out of the eight analysed use cases showing prioritisation both in policies and insurance market.

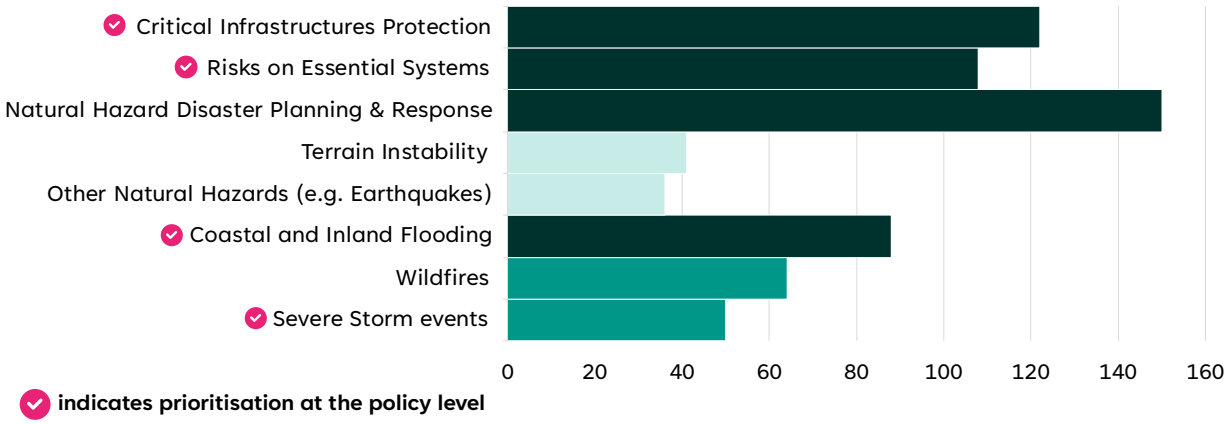


Figure 19: Distribution of Earth Observation projects across Peace & Security and Climate Resilience & Emergency Management use cases related to the defined Finance & Insurance market segments, categorised using a three-tier scale

In the context of project distribution, while this is not uniform across the use cases shown in the graph, the lower levels of projects are still relatively high. This demonstrates that use cases relevant for the insurance domain are generally well addressed by Earth Observation projects, though Earth Observation project uptake for geological hazards (e.g. earthquakes, landslides) remains lower. Therefore, where Earth Observation already serves as a primary tool for monitoring climate impacts or infrastructures, there is potential for it to deliver further benefits if leveraged beyond this traditional role by actively supporting risk reduction and more operational activities.

More **specific opportunities** can be identified in the following areas:

- *Critical Infrastructure & Essential Systems, esp. Food* – These use cases show strong alignment between policy priorities and insurance market coverage. Opportunities lie in Earth Observation transitioning beyond a monitoring role towards operational solutions. For instance, Earth Observation can be leveraged to support insurers in verifying that post-disaster reconstruction meets resilience and sustainability standards.
- *Water-related and Atmospheric Hazards, e.g. Flooding & Storms* – Highly prioritised in policies, these use cases are significantly covered by insurance products (above 80% saturation) and show a certain level of maturity in terms of Earth Observation uptake. Major opportunities lie in further integrating Earth Observation into advance risk assessments and further supporting the insurance industry in shifting beyond reactive payouts towards preventive action.
- *Wildfires* – Despite high prioritisation at the policy level, especially for Southern European countries, wildfire coverage is more limited in the insurance market. Earth Observation is uniquely positioned to bridge this gap, already delivering significant benefits for wildfires, particularly through thermal imaging capabilities. Near-real-time thermal infrared capabilities can enable the development of parametric insurance, where automatic payouts can be triggered by physical parameters verified by Earth Observation (e.g. fire intensity and heat signatures), ensuring immediate liquidity for rapid response and recovery.
- *Geological Hazards* – Though often secondary in climate resilience policies and showing limited Earth Observation project uptake, terrain instability and landslides represent a relevant pillar of the insurance market (over 60% coverage for Property). The alignment gap between insurance and policy interest represents an opportunity for increased public-private collaboration, where insurance risk frameworks can guide public authorities towards high-priority and risk-prone areas, enabling a targeted use of Earth Observation to inform more resilient urban and rural planning.
- *Wars, Riots, and Civil Unrest* – Limited insurance coverage for this rising threat (20% for Property; 11% for Motor). To bridge this gap, Earth Observation can be leveraged as a source of independent, verifiable data to support remote damage assessments in regions where on-the-ground access is limited.

8 INTEGRATION OF SPACE INTO POLICIES

Beyond the concrete use of Earth Observation to serve policy priorities (and private interests), a key aspect for assessing and boosting the policy impact of Earth Observation concerns its integration into policy and strategy documents. This can also serve as an indicator of the level of awareness among policymakers regarding the use and potential of space, specifically Earth Observation, for achieving their objectives. Indeed, sectoral policies typically list or refer to the primary tools for their implementation and monitoring, as well as briefly mentioning the technologies used for assessing the context and formulating policy objectives and subsequent actions. **Explicit references to a specific technology or its direct applications can facilitate and streamline budget allocation for the development of enabled services**, while also setting a clearer and more stable (i.e. trusted in the longer term and beyond political cycles) list of needs to be met by the industrial ecosystem.

Once a list of terms univocally related to Earth Observation was defined (*hereafter* 'dictionary'), all collected sectoral policies were screened for occurrences of the words contained in the dictionary (for further details See Annex A – Extended Methodology). The results are provided both in absolute number of mentions (Figure 20) and normalised by the number of pages of the screened documents (Figure 21), which typically differs across domains. Overall, this exercise showed **varying levels of integration of Earth Observation across the domains within the analysed sectoral policies**. While the climate and agriculture domains stand out with the highest levels of integration, there is a clear potential for increase in the other domains, especially Energy and Peace & Security.

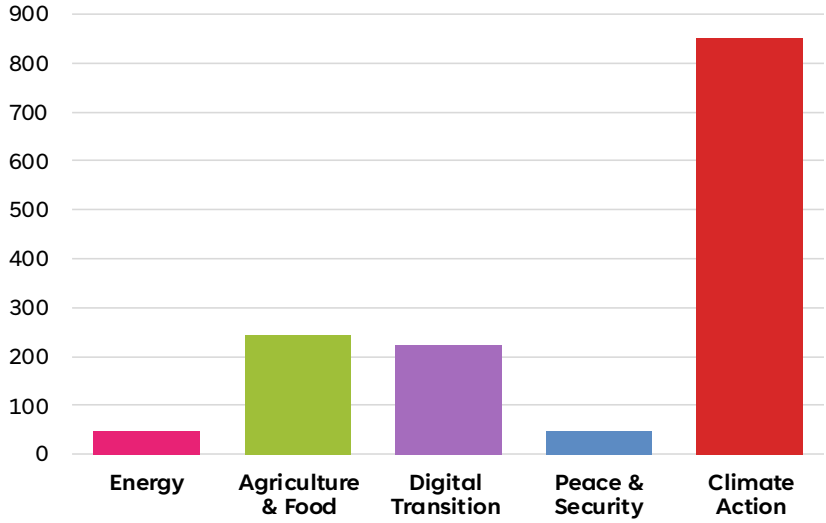


Figure 20: References to Earth Observation in policies (absolute numbers)

These results include only references clearly linked to Earth Observation solutions, also taking into account the terminology specific to certain domains (e.g. satellite surveillance or space-based ISR in security policies). Nevertheless, non-explicit mentions of functions frequently enabled by Earth Observation (e.g. 'Monitoring' or 'Reconnaissance', respectively in the climate and defence domains) may also indicate underlying integration of Earth Observation in policies and represent entry points for increasing uptake.

In particular, several domain-specific insights can be derived from Figure 20 and Figure 21.

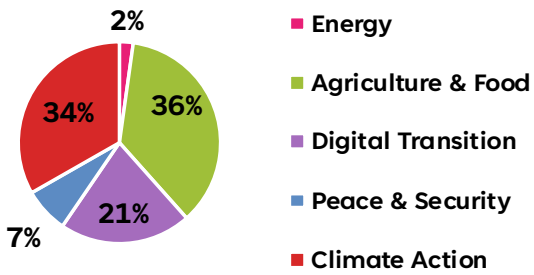


Figure 21: References to Earth Observation in policies (figures normalised by length of documents)

8.1 Climate action

References to Earth Observation in climate policies, also due to the extensive work conducted by ESA, the EU and national agencies to develop services and awareness of their value in this domain, should be understood as covering not only climate resilience, but also climate mitigation. This is due to the strong interconnection of these two components of climate action within the analysed policy texts. The high number of Earth Observation mentions, including when normalised to account for the significant length of these policies, seems well aligned with the broad range of projects addressing climate resilience use cases (See §6.2).

This high number of references to Earth Observation in climate policies is distributed across the analysed countries as shown in Figure 23. Among the countries with higher levels of integration of Earth Observation, Italy, the UK and Greece show substantial alignment between broader climate policies (National Communications, NCs) and more specific climate resilience policies (National Adaptation Plans, NAPs), with significant references to Earth Observation in both documents. Germany, however, shows more integration of Earth Observation in its NC, thus signalling a **more prominent use of Earth Observation for mitigation than for resilience**. Overall, the average number of mentions per country is relatively high, with some notable exceptions, such as France.

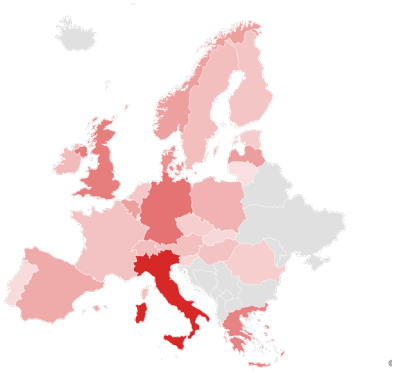


Figure 23: Countries by number of references to EO in climate policies (darker equals higher number)

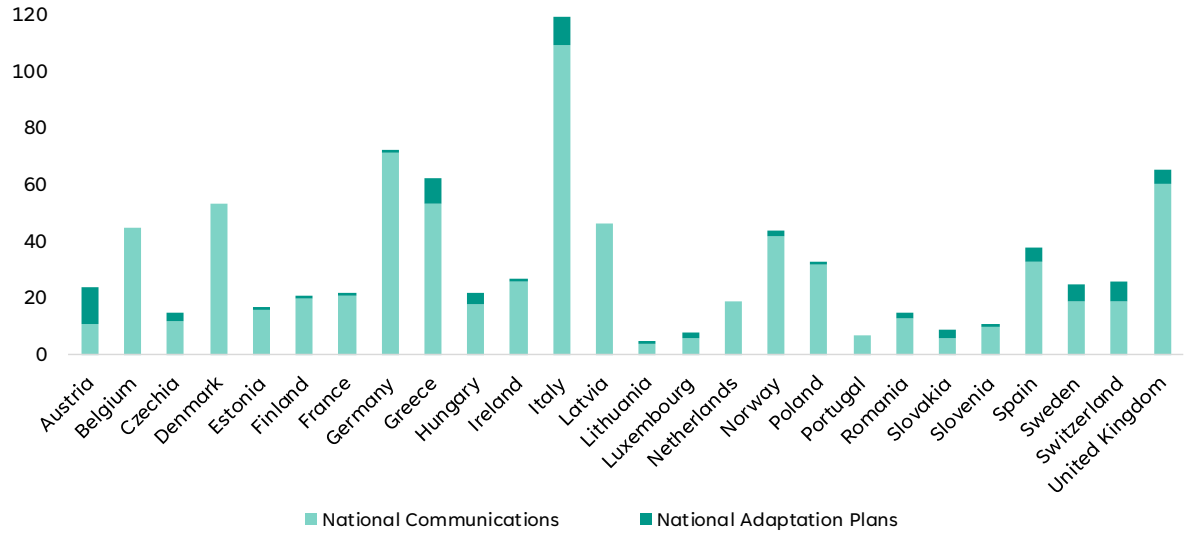


Figure 22: Number of references to Earth Observation in NCs & NAPs by country

When looking more specifically at the terms (related to Earth Observation) mentioned in the reviewed documents, references to the Copernicus programme and its Sentinel satellites account for more than a quarter of all Earth Observation mentions. Italy, Germany and Greece, in particular, primarily refer to Copernicus in their policies, with the latter also significantly recalling the role of EUMETSAT. In contrast, the UK tends to include more generic references to Earth Observation in the analysed documents.

8.2 Agriculture & Food

Agriculture policies also show a good level of integration of Earth Observation, especially considering the normalised data in Figure 21. In this context, national implementation strategies of the EU Common Agriculture Policy – where available – especially mention Earth Observation in their section on monitoring and modernisation, including knowledge, innovation and digitalisation. While this aligns with the current significant use of Earth Observation for agriculture, the comparison of the total number of references to Earth Observation with the climate domain (Figure 20) may reflect the less tangible value currently delivered to end-users, notably farmers. This represents an opportunity to develop services capable of impact at the local level while ensuring scalability (See §3.3), building on the demonstrated added value of Earth Observation in the sector. In turn, this would require dedicated budget, more easily justified should agriculture and food policies and strategies reference Earth Observation more frequently in their text.

Mentions of Earth Observation show a broad distribution across most European countries, with Italy leading, together with Slovenia, particularly in the national implementation strategies of the EU Common Agriculture Policy and, where available, food security strategies. When comparing the results with those for climate resilience, also considering that climate-induced events can significantly affect agricultural activities, a reduction in mentions is observed for Germany and the UK, with stable results for France, Greece, Belgium and Austria.

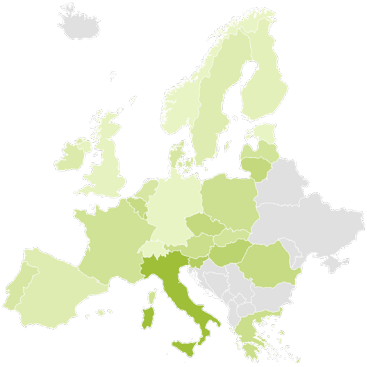


Figure 24: Countries by number of references to EO in food & agriculture policies (darker equals higher number)

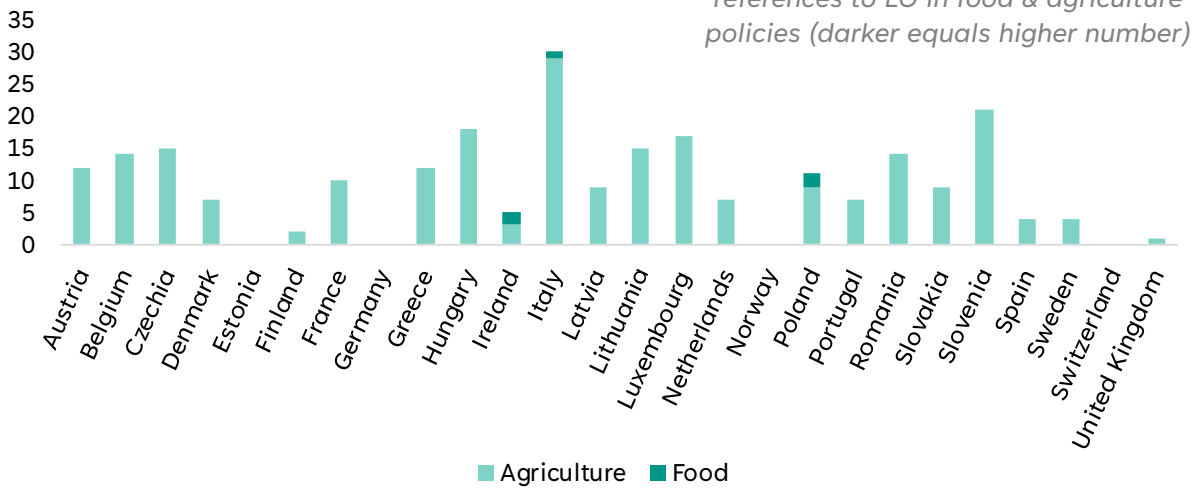


Figure 25: Number of references to Earth Observation in agriculture and food security policies by country

While references to Copernicus and its Sentinels are still present, the terms privileged in agricultural policies are more generic, referring to 'Remote Sensing' or 'Geospatial' data and technologies. This could signal a less pronounced awareness of the specific benefits of programmes such as Copernicus for the agrifood sector, as well as a more limited direct impact of the latter (See also in §8). In addition, these more generic references could indicate a need for more integrated technologies for supporting agriculture from above, not limited to Earth Observation. When considering the distribution of main terms across countries, instead, new entrants emerge compared to the climate policy case.

8.3 Peace & Security

Security and defence strategies make limited reference to Earth Observation, even when accounting for domain-specific terminology and normalising data considering their shorter length compared to climate policies, for instance. This fact contrasts with a significant number of Earth Observation projects being relevant for security applications, especially in the civil security, emergency management and law enforcement domains – with defence projects more difficult to review due to the limited transparency (See further in §5). This fact suggests both demonstrated operational value and untapped potential of Earth Observation for security applications, still to be fully acknowledged at the policy level.

The distribution of Earth Observation references appears relatively uniform across European countries, with higher levels noted for Finland and Luxemburg. In many cases, countries refer to Earth Observation in their policies without citing specific programmes, instead using terms such as 'Geospatial' data and technologies in support of civil security and military operations. Moreover, almost half of the references appear in connection with security-specific terms such as 'ISR' or 'Reconnaissance'. Finally, several mentions are also linked to ensuring the security of space infrastructure, including Earth Observation satellites.

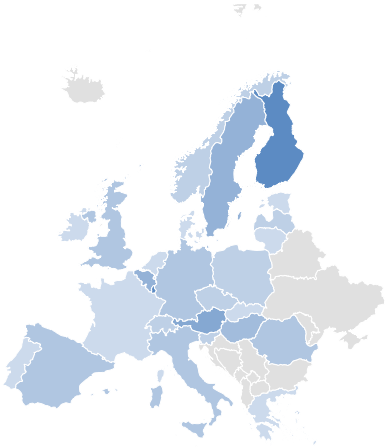


Figure 26: Countries by number of references to Earth Observation in security policies (darker equals

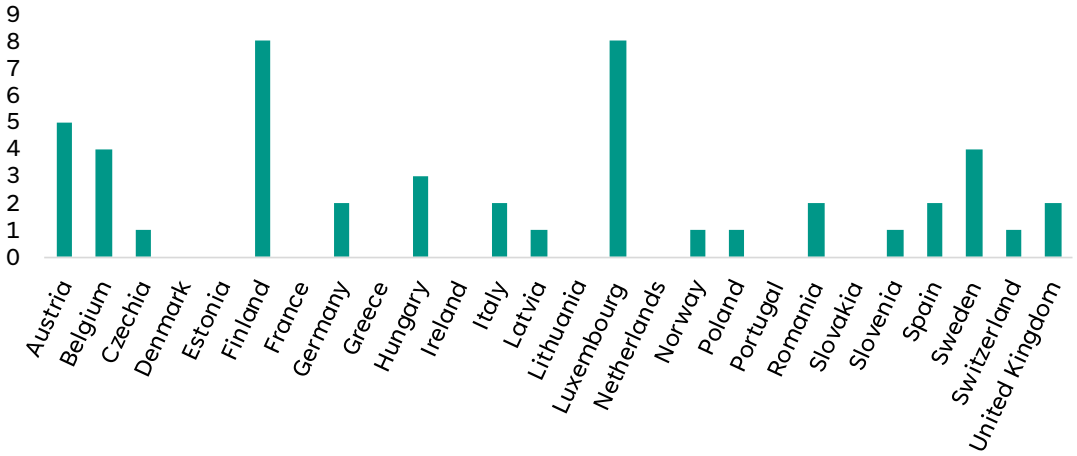


Figure 27: Number of references to Earth Observation in security, defence & foreign affairs policies by country

8.4 Energy

The energy policies analysed contain very rare references to Earth Observation. While experts in the field confirm that links between space and energy are largely absent across all energy-related policies, the limited integration of Earth Observation in National Energy and Climate Plans is especially notable because of the significant length and nature of these documents, at the intersection of the energy and climate domains and their role as key reference for the energy transition in Europe. Other energy policies reviewed provide even fewer mentions. This aligns with the limited number of Earth Observation projects specifically addressing energy-related use cases, with the exception of infrastructure monitoring and, to a lesser extent, deployment of renewables.

The limited integration of space in the energy sector signals an opportunity to expand the use and benefits of Earth Observation in this domain. Higher awareness and broader integration of Earth Observation into policies can stimulate the development of new and more tailored Earth Observation services. In practical terms, references to Earth Observation can be introduced particularly in the initial sections of policy documents, mentioning how Earth Observation solutions are already contributing with proven benefits to activities in the domain. References should also be included when listing the tools for implementing the policy measures; these would represent a starting point for developing more precise implementation plans specifying the opportunities and constraints on the use of Earth Observation for specific use cases.

The low level of integration of Earth Observation within the analysed documents is relatively uniform across countries, with the exception of Slovenia and, in particular, Greece.

References to Earth Observation in the case of Greece and Slovenia mainly include 'Geospatial' technology and 'GIS' data and services, especially for carbon removal, emission reduction, air quality control, deployment of renewables, monitoring of energy infrastructure and estimating energy consumption.

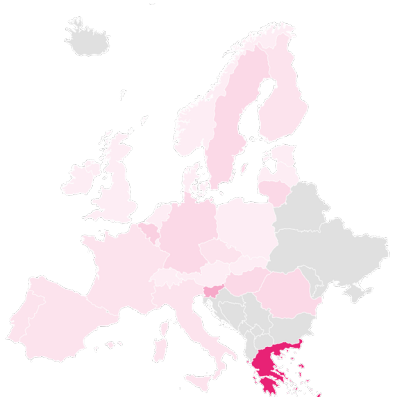


Figure 28: Countries by number of references to Earth Observation in energy policies (darker equals higher number)

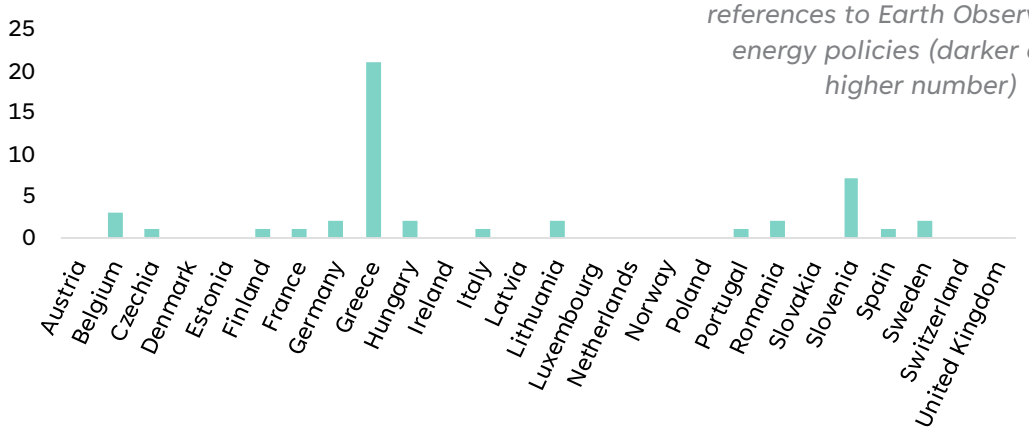


Figure 29: Number of references to Earth Observation in National Energy and Climate Plans by country

8.5 Digital Transition

Digital transition policies show a good level of integration of Earth Observation, which nevertheless requires interpretation. These policies refer to Earth Observation both as an enabler of digital solutions through the provision of large volumes of data (e.g. feeding AI-based tools), and as one of the technologies to be developed to support decision making and policy implementation in other sectors (e.g. agriculture). This second 'type' of references – activities that enable Earth Observation in various sectors more than other digital services or tools – represents the majority of mentions and aligns with the outcome of the project analysis. Indeed, the latter shows a prevalence of Earth Observation solutions for enhancing real-time assessing and decision making (in various policy domains), outweighing by far the number of projects aimed at enabling digital services or supporting other priorities of the digital transition (e.g. energy efficiency of digital infrastructure). This indicates untapped potential to further integrate

Earth Observation into digital transition strategies, particularly by emphasising its role as a data provider for digital service delivery.

The integration of Earth Observation in digital policies is not uniform across the selected countries, with about half of them including a baseline level of references to Earth Observation in the policies, with two notable exceptions in the case of Greece and Latvia. Overall, many terms related to Earth Observation are mentioned in the analysed policies, from 'Copernicus' and 'Sentinel' to more generically 'Earth Observation' or 'Remote Sensing'. 'Geospatial' data and technologies record the highest number of occurrences within the dictionary.

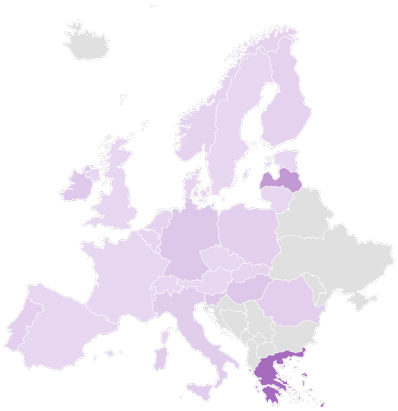


Figure 30: Countries by number of references to Earth Observation in digital policies (darker equals higher number)

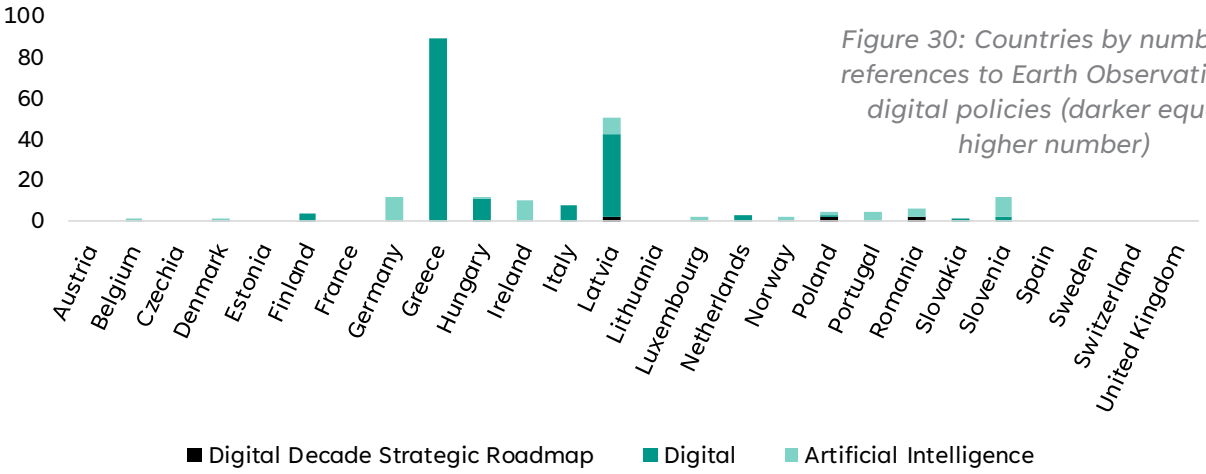


Figure 31: Number of references to Earth Observation in Digital and broad AI strategies by country

Overall, this analysis of the integration of Earth Observation into policies highlights:

- The level of awareness among policymakers on the role and value of Earth Observation in achieving their objectives and the ones that Earth Observation is most closely linked to.
- Successful national experiences that may serve as a reference for further integration of Earth Observation.
- The terminology most commonly used to refer to Earth Observation in policy documents, accounting for differences across various policy domains.

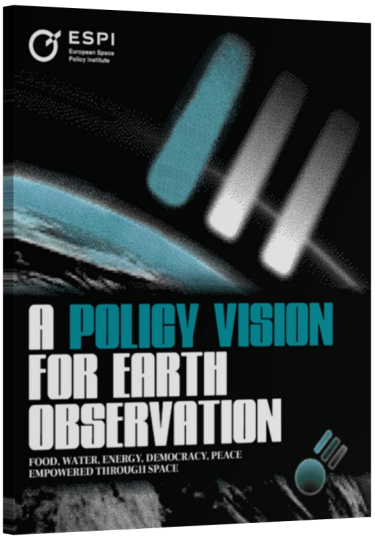
While this analysis can stimulate actionable considerations and potentially activate efforts to address the gaps identified, it does not constitute a full assessment of the integration of Earth Observation in the selected domains. Nevertheless, it provides a solid foundation for further investigations (e.g. through extensive consultations with various policy communities) to assess awareness and capacity of policymakers, including operational units in public administrations, to leverage existing solutions and inform the development of new ones. Further research at the level of sub-sectoral policies (e.g. solar energy policy within the energy domain, or wildfire management strategy as part of climate resilience) would also be valuable, given that most policy implementation plans – and the related identification of the tools for implementation – are developed at this sub-sectoral level.

9 NEXT STEPS

The European Space Policy Institute (ESPI) is taking action on the outcomes of this study, contributing to embedding Earth Observation at the core of Europe's policies.

Three main activity lines are currently being pursued:

1. Active collaboration with policymakers in selected domains, stemming from the release of A **Policy Vision** for Earth Observation, published in June 2025 in collaboration with the European Space Agency (ESA), and reinforced by the analysis herein. ESPI aims to strengthen the understanding of how innovation is embedded into policymaking, leading Europe to widespread uptake and benefits of Earth Observation solutions across different policy sectors.

   Beyond existing collaborations, ESPI invites non-space experts in key fields to contribute, bringing their knowledge to better integrate space technology, especially Earth Observation, into European and national policy cycles.

2. An independently funded project focusing on the intersection of space, water security, water management and water diplomacy, with the final aim of contributing to the co-definition and co-implementation of a joint **Space & Water** policy agenda. Water-related objectives are explored across international, European and national policies, while the impact of space within will be assessed across all applications. Key trends at the interface of Space and Water policies will be identified, and recommendations for coordinated action will be formulated.

   Further efforts will concern the **Energy** domain, in light of the findings of this report.

3. Finally, building on input from policy communities (1) and sector-specific research (2), ESPI will work to align needs with new funding and existing programmes. This means identifying opportunities in both publicly funded initiatives and market-driven projects, such as business-to-business partnerships, public-private collaborations, and new activities under European flagship programmes.

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This report could not come to life without the continuous support and constructive exchange with the personnel of the European Space Agency (ESA), efficiently coordinated by Lucie Lachkar, Policy Officer in the Strategy, Programme & Coordination Office of the Directorate for Earth Observation Programmes at ESA. Our gratitude also extends to the Director of Earth Observation Programmes at ESA, Simonetta Cheli and other ESA colleagues contributing to the conceptualisation, development and review of this document, notably Rune Floberghagen, Eleni Paliouras, Inge Jonckheere, Giuseppe Ottavianelli, Alessandra Tassa, and Sascha Deutsch.

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ANNEXES

Annex A – Extended Methodology

The study's methodology encompasses multiple stages, comprising data collection, analysis, validation, and synthesis steps. These are outlined below, beginning with an overview of the scope and the initial selection of domains.

Geographical and Temporal Scope

The study focuses on the 26 Member States (MS) and Associate Member States (AMS) of ESA as of 2025; Austria, Belgium, the Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, the Netherlands, Norway, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden, Switzerland, the United Kingdom.

The temporal scope, impacting both the policy and project collection, was defined as follows:

- For the policy collection activity, it could differ depending on the domain, but in general, priority was given to the most recent documents available for each domain.
- For the project collection activity, the team only collected projects that were ongoing or planned at the time of collection (final review in Q3 2025).

Domain Selection

Six domains were selected for the subsequent policy and project analysis. A detailed down-selection procedure was conducted from an initial selection of seventeen domains, listed in Figure 32. These were identified through several sources and methods aiming to develop a comprehensive list, including screening domains addressed by national ministries, AI tools, ESPI databases and previous policy-related work. The results were also benchmarked against the EU space market segments.[34]

Four down-selection criteria were developed to align with the study's primary aim, i.e. to examine how current Earth Observation solutions support Europe's most urgent policy objectives and to identify opportunities to expand their uptake in these areas The criteria are prioritised in this order:

- The identification of the top short-term national priorities of ESA MS and AMS
- The identification of the top short-term priorities of Europe, incl. EU, more generally
- An assessment of current and future economic domain potential through global Earth Observation revenue by market segments

To identify short-term priorities for ESA MS and AMS, extensive desk research was conducted across a range of documents, notably governmental summaries of speeches, institutional strategies and documents, policy statements & governmental declarations at the beginning of new governments (incl. objective setting), and national recovery plan priorities. The goal was to identify short-term (since the most recent government took over, or in the last four years), high-level priority objectives, i.e. those identified by the government as their top pressing or strategic

[34] EU Space Market Segments - EUSPA

priorities. The type of documents analysed remained consistent across all countries to ensure the analysis remained solid and appropriately high-level.



Figure 32: Initially selected domains

The outcome of the activity was a list of priorities which allowed to develop the list of domains shown in Figure 32. To this end, similar items were grouped together into one domain, e.g. the objective to gradually replace heating systems powered by fossil fuels and the acceleration of decentralised, renewable energy generation and resilient electricity grids were grouped under the Energy domain.

Subsequently, a calculation of the number of countries prioritising each of the domains was pursued. A country was considered to prioritise a domain when it would have addressed a number of priorities within such a domain above a certain threshold.

In addition to national high-level priorities, the other two criteria were fulfilled as follows:

- European Short-Term Priorities

  European short-term priorities were analysed through the consultation of many resources, including through the assessment of European Union and European Commission priorities, such as the European Commission 2021-2027 Budget Priorities.[35] These priorities were also benchmarked against the Copernicus for EU Green Deal and other EU Policies, as well as

[35] 'European Union priorities 2024-2029'; 'Commission's Priorities'; '2021-2027 long-term EU budget & NextGenerationEU' - European Commission; 'The 17 Goals' – United Nations

international references such as the United Nations Sustainable Development Goals (SDGs).[36] The importance of the priorities was deemed to be low, medium, or high based on the urgency assigned, and frequency of mention within the aforementioned resources.

- Global Earth Observation Revenue by Market Segments 2023 & 2032

  Two resources were used to investigate Earth Observation revenues in 2023 and 2032: EUSPA's Earth Observation and GNSS Market Report and New Space Portugal's Earth Observation Market Segments data.[37] In a few rare cases (Education & Training, Diplomacy & International Cooperation, and Water Resource Management), there was no equivalent domain in these sources to the ones selected, but this was accounted for in the final down-selection step (outlined in Figure 33). Domains were deemed to have low, medium, or high revenue based on their percentage of market revenue (low: <5%, medium >5%, <10%, high: >5%).

The final down-selection step involved aggregating all results and converting them into a points system to determine the top domains. An example of this for two domains is shown in Figure 33.

	Top National Priorities (26)	Short-Term Priorities for Europe (20)	Global Market Segments 2032 (15)	Global Market Segments 2023 (10)	Points
Digital Transformation	23/26	High (20)	N/A	N/A	43/46
Health & Access to Care	18/26	High (20)	Low (5)	Low (3.33)	46.3/71

Figure 33: An example of the final down-selection step, used to create a ranking for each domain

National ESA MS and AMS priorities were deemed to be most important and were assigned a total points value of 26 (for each country prioritising that domain), short term priorities were assigned a total points value of 20 (low: 6.66, medium: 13.33, high: 20), 15 for market segments in 2032, and 10 for market segments in 2023. In the aforementioned three cases where no market segment information was available, the ranking for these domains was calculated using only the first two criteria. The ten top ranked domains are shown in Figure 34.



Figure 34: Top ranked domains

[36] 'Supporting the European Green Deal' – European Commission
[37] 'EUSPA Earth Observation and GNSS Market Report' (2024) – EUSPA; 'Role and Uses of Earth Observation Data' – New Space Portugal

Final domains were aggregated based on overlaps and complementarity, to reach the final six domains shown in §1 and addressed in the subsequent research.

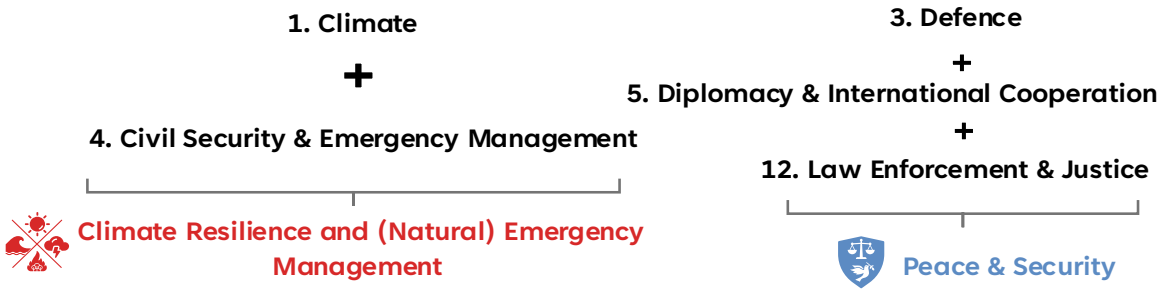


Figure 35: An example of final domain aggregations

Policy Collection & Analysis

Policy documents were collected for all selected domains through a combination of desktop research, the use of external datasets such as FAOLEX and OECD, consultations with relevant stakeholders, and through the aid of data science search tools. The particular case of Finance & Insurance is further explained in the following section. Over 300 national policy documents were collected, excluding policies addressing niche areas within a domain (e.g. organic farming). The types of documents collected for each domain are shown in Figure 36, with priority given to the latest policy documents available for each domain.



Figure 36: Policy documents collected for each domain

Three approaches were combined to **identify use cases** for each domain:

- A bottom-up approach, whereby priority use cases across domains were directly identified within the policy documents via inter alia frequency of mention or explicit reference to specific measures being adopted to address particular objectives and mention of urgency.
- A top-down approach, whereby priority use cases were identified via desktop research and external resources such as recent high-level reports dealing with the analysed domain.
- A consultation campaign with policymakers within each domain to refine, validate and integrate the selected use cases.

While the **Finance & Insurance** domain has been selected among the prioritised areas for policymakers, its analysis required the development of a dedicated methodological approach (See §7). In the absence of dedicated policy documents, the decision was taken to analyse the insurance market in the countries within the scope of the study. While this represents only a small section of the finance domain, it is characterised by sufficient uniformity to allow quantitative analysis on the interests pursued by insurance companies and its comparison with the public objectives in domains of policy such as the ones mentioned above.

The analysis of the insurance segment entailed the selection of four non-life insurance companies per country, two of which were selected by highest revenues and two with the additional criterion of having their headquarter in the country. This allowed the team to encompass both multinational companies with their local subsidiary and other companies mainly active only on the national territory. The manual selection was complemented by AI-driven results validation. Life insurance companies were excluded from the analysis due to their focus on specific products with limited interest for the other policy domains analysed, producing an overall bias on the results. This accounted for a total of 104 insurance companies screened.

Main insurance-derived products were defined and identified, based on the products of the companies analysed. All identified products chosen for the selected companies were reviewed and categorised within the following preset list:

- **Private** – Property and Motor
- **Business** – Agriculture, Construction, Transport & Logistics

This allowed the identification of the main risks insured across these categories and sectors.

Furthermore, the outcomes of the analysis were matched with the policy analysis in the other domains studied, in particular highlighting those use cases attracting both interest of insurance companies and representing a priority for sectoral policymaking (e.g. floods).

Finally, the statistics on projects addressing the use cases characterised by this convergence of public and private interests were aggregated and produced in a single graph (See §7.2), building on the data collected for the other domains analysed. Indeed, most of these projects could serve the interests of insurance companies, while not being primarily aimed at their use.

Project Collection & Analysis

The collected projects include:

- **Research Projects:** Generally, time-limited, specific initiatives that are often led by public and academic institutions. For instance, feasibility projects. This was chosen if the result of the project is or will be a research paper.
- **Tools/Platforms:** An online system or application that incorporates space data and provides a specific service or tool. These systems can include online databases and data repositories, as well as commercial products that use space data, amongst others. This was chosen if the result of the project is or will be a tool or platform.
- **Programmes:** A wider initiative that involves several activities and objectives. Activities in this category may include capacity-building and training programmes.
- **Others:** Projects that feature unique qualities and are not characterised by the same specificity as the above-mentioned categories.

Although upstream projects such as single satellites and constellations were also collected, they were not included in the final analysis, to ensure consistency among the projects compared. Projects were only included if Earth Observation was specifically being used to address those use cases. Collected projects were added to an internal database, with the information provided in Table 1.

Table 1: Project details collected

Project Details Collected	Description
Region of the managing entity	The region of the entity managing the project If there is more than one contractor and they are all from the same continent, that continent was chosen, i.e. Europe, Asia, Africa, etc. "International" was chosen as the managing region if there are many contractors involved from different continents. **European sub-regions**[38] **Northern Europe** Denmark Estonia Finland Ireland Latvia Lithuania Norway Sweden United Kingdom **Western Europe** Austria Belgium France Germany Luxembourg Netherlands Switzerland **Southern Europe** Greece Italy Portugal Slovenia Spain **Eastern Europe** Czechia Hungary Poland Romania Slovakia
Region of impact	The region over which the project is being used or implemented National/Continental/International (See above). The Arctic and Antarctica are also included.
Scale of implementation	Local/National/Continental/Global
Entities involved	"Yes" if more than one participating entity involved in the project
Project Status	Ongoing/Planned

[38] 'The Geoscheme of the United Nations' - WorldAtlas

Examples

To clarify the above table, a few examples are proposed:

- A project managed by an entity based in the United Kingdom to develop a Digital Twin in Athens:
  - Region of the Managing Entity: United Kingdom
  - Region of Impact: National (Greece)
  - Scale of Implementation: Local
- A project managed by a European entity with the aim to develop an online platform/tool for agricultural monitoring:
  - Region of the Managing Entity: Europe
  - Region of Impact: International
  - Scale of Implementation: Global
- A project managed by an entity based in Greece with the aim to address wildfires in four national parks across Greece, Italy, Spain, and Portugal:
  - Region of the Managing Entity: Greece
  - Region of impact: Continental
  - Scale of Implementation: Local

Projects were collected via an extensive desktop search, also involving the use of AI and general search tools to expand the list of sources. Projects were searched for on a country-by-country basis using keywords related to Earth Observation, and domain-specific terms. Usually, a single search would reveal further links to additional Earth Observation projects of interest. When all available links and search terms had been exhausted for a particular country, the search would resume for the next one in the list. National project and news repositories were consulted, as well as most recent reports of various organisations active in the field. Additional projects were found in already existing project databases, e.g. the eoPortal.[39] Industry databases and resources were also used to aid the search for projects, such as ESPI's internal Space Venture database. While systematically screening this set of sources to maximise comprehensiveness, this remains nevertheless limited by several constraints (e.g. language), in particular for projects with very localised or niche impacts.

For each project, its objectives were identified as explicitly mentioned within project descriptions and matched with the relevant use cases within the selected policy domains. All project objectives were identified, resulting in each project potentially being relevant for more than one use case, including across multiple domains.

Opportunities Identification

The number of countries prioritising each use case within their policies (*hereafter* level of prioritisation) was compared to the number of projects addressing such use case (*hereafter* level of contribution of Earth Observation solutions). The level of prioritisation is represented through

[39] 'Satellite Missions catalogue' - eoPortal

a 5-tier threshold system, based on the share of countries (e.g. top level for objectives prioritised by over 90% of analysed countries). The level of contribution of Earth Observation solutions is instead displayed through a 3-tier system, with a rescaling of the maximum threshold by domain, with domains being addressed by a significantly higher number of projects than others.

This resulted in the identification of potential gaps, represented by the use cases with a high level of prioritisation and low level of contribution of Earth Observation solutions.

Impact Assessment Case Studies

The policy impact assessment case studies aimed to evaluate the benefits of Copernicus data integrated within the operations and decision-making processes of national and European public administrations (PAs). In line with this objective, the following activities were conducted:

- Selection of Areas of Investigation

  The research and consultation focused on three policy areas in line with the six selected policy domains, ensuring significant diversity among them to capture a range of sectoral dynamics, including market structures and governance models:

  - **Maritime** – Focus on security, maritime safety, and environmental protection (e.g. oil spill detection and Illegal, Unreported, and Unregulated (IUU) fishing), with commercial interests driving significant private-sector involvement.
  - **Agriculture** – Wide variety of applications and strong emphasis on compliance monitoring (e.g. the Common Agricultural Policy (CAP), Land Parcel Identification System (LPIS) updates, and land-use verification).
  - **Forestry** – Centered around forest preservation and sustainable resource management, largely shaped by public policies and actors.

  More specifically, the selection of the areas was based on**:**

  - The significant uptake of Earth Observation in the activities conducted within the identified area, which guarantees the necessary data availability to support the impact assessment.
  - The willingness of at least one of the engaged PAs operating in this area to provide data and be consulted as part of the study.
  - The alignment of the area with the six overarching policy domains, thus connecting the impact assessment with the policy priorities of European countries.

- Identification of Public Administrations and Case Study Selection

  Within the three areas, **five case studies** were developed in collaboration with five PAs:

  - Maritime: Spanish Civil Guard (Guardia Civil) and European Maritime Safety Agency (EMSA).
  - Agriculture: Slovenian Agriculture Paying Agency (ARSKTRP) and Italian Agency for Agricultural Payments (AGEA).
  - Forestry: Czech Forestry Institute (NLI).

  PAs were selected based on three main criteria:

  - Level of their adoption of Copernicus data and the presence of ongoing Earth Observation projects, ensuring the selection of case studies representing different levels of adoption.

- Availability of established contacts and trusted relationships with ESPI and ESA.
- Willingness to share data and actively participate in consultations was considered an essential prerequisite.

Overall, the selected entities demonstrated a strong commitment to dialogue and collaboration, which ensured the collection of reliable empirical evidence and enabled the assessment of Earth Observation impacts.

- Data Collection and Analysis

  The collection of data was based on both qualitative and quantitative approaches, initially involving desk research, including websites of public administrations and ESA, as well as analysis of relevant documents. The primary information was then gathered through a combination of exploratory interviews with open-ended questions to enable a deeper understanding of the current level of satellite data use, followed by tailored questionnaires. The questionnaires combined questions with predefined ranges and a limited number of open-ended questions, enabling both quantitative and qualitative insights. To quantify the impact of Copernicus data, a set of Key Performance Indicators (KPIs) were defined in collaboration with each PA and grounded in concrete examples discussed during the interviews. These KPIs covered dimensions such as cost and time savings, levels of innovation and digitalisation, and social and environmental benefits. Finally, the results were compiled into one-page factsheets on each use case, showing examples of 'Uptake in Practice', and validated by the entities consulted.

Policy Text Extraction

The analysis of the integration of Earth Observation in the text of policy documents was conducted through text extraction methods developed in Python, such as Natural Language Processing (NLP) and document processing. The NLP model used in this case is the spaCy model en_core_web_sm, a lightweight pipeline trained on web text. Specifically, this was used to find any reference to 65 Earth Observation keywords in the policy documents, i.e. the Earth Observation dictionary. The preprocessing step included cleaning the documents (e.g. removing extra whitespace or correcting unintentionally split words).

The dictionary is an ESPI-collated list of words related to Earth Observation, including basic terms such as "Earth Observation" itself and "remote sensing", technical terms such as "lidar" and "thermal imaging", and specific satellites, programmes, and systems such as "Copernicus" and "Landsat". For the Peace & Security domain, the dictionary was broadened to capture space uses that may be referenced without explicitly mentioning Earth Observation, such as "intelligence" and "reconnaissance", which are frequently used in this context. Only references to space-based intelligence or reconnaissance capabilities were filtered and counted. Manual checks of all occurrences were also undertaken as an additional verification step. Once the references were extracted, overall references per domain were normalised to the number of pages in all policies analysed for each domain by dividing the total number of pages by the total number of references.

For the case of Finance & Insurance, due to the absence of dedicated policy documents, the analysis on the number of references to Earth Observation was not conducted, and no particular mention of space was found in the documents of insurance companies, also due to the limited external transparency concerning the tools used to track and monitor insured risks.



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